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List of Abbreviations

CT - Computed tomography

DLSE - Deep-learning-based scatter estimation

EBSE - Energy-based scatter estimation

FDG - ^{18}F-fludeoxyglucose

FOV - Field of view

FWHM - Full width at half maximum

LOR - Line of response

MoM - Method of moments

MR - Magnetic Resonance

PET - Positron emission tomography

PDF - Probability density function

PHG - The Photon History Generator module

SimSET - Simulation System for Emission Tomography

SIRF - Synergistic Image Reconstruction Framework

SSS - Single scatter simulation

STIR - Software for Tomographic Image Reconstruction

TOF - Time-of-flight

1. Abstract

Photon scattering leads to bias and artefacts in reconstructed positron emission tomography (PET) images, resulting in risks of misdiagnosis. A widely used approach in clinical scatter correction, single scatter simulation (SSS), cannot provide accurate modelling of the complex scattering in whole-body scans. Energy-based scatter estimation is an alternative that avoids many limitations of SSS, as it focuses on the difference between unscattered and scattered photons' energy information to estimate the scatter distribution more accurately. In this work, one energy-based scatter estimation approach was evaluated using Monte Carlo simulations for more realistic physical modelling of photon interactions. The results of our method were compared with the ground truth from the simulation to assess scatter estimation accuracy. Our investigation has demonstrated that the application of energy-based scatter estimation approximates unscattered photons with high precision, but it requires further calibration to accurately model scattered photons before extending its potential to clinical implementation.

2. Introduction

2.1 Positron Emission Tomography

Imaging is a key process that guides and monitors diagnostics and therapeutics in clinical research. Positron emission tomography (PET) is a widely used imaging modality with high sensitivity and specificity across a wide range of physiological and pathological processes (1). Various studies have demonstrated the diagnostic potential of PET in oncological, paediatric, cardiac, and neurological imaging in both research and clinical settings (2-4). PET enables the absolute quantification of metabolic processes and the measurement of functional metrics by tracking the spatial distribution of radiotracers with extremely high sensitivity; only trace amounts of radiotracers in the picomole range are necessary for imaging without significant adverse radioactive effects (5, 6). PET relies on cyclotron-generated, positron-emitting radiotracers such as Fluorine-18, which can be used to produce ^{18}F-fluorodeoxyglucose (^{18}F-FDG), a glucose analogue. FDG is commonly used for monitoring glucose uptake and metabolism (6, 7) and, consequently, multiple cellular processes and functional abnormalities. It is particularly valuable in assessing tumour metabolism and relapse in oncology research, as well as global and regional brain glucose consumption (3). The visualisation of functional processes makes PET outstanding among different imaging techniques.

Radioactive decay is the fundamental principle of PET. When a positron-emitting radiotracer is injected, it undergoes β^+ decay, where one proton in the proton-rich nucleus is converted into a neutron, emitting a neutrino and a positron, the antimatter counterpart of the electron. The annihilation of a positron with an electron emits two γ-photons in opposite directions, each with an energy of 511 keV. (4, 5, 8-11). The photons are emitted isotropically, meaning that the probability of emission at all angles is uniform. PET detectors are arranged in a ring configuration, where the detection of two γ-photons in coincidence defines a line of response (LOR) (6, 10), as shown in Fig. 2.1A. Conventional PET does not precisely measure the difference between the arrival times of the two photons, resulting in a uniform probability distribution of the emission location along the LOR (Fig. 2.1B). This uncertainty causes noise correlation when LORs from different annihilation events intersect (Supplementary Fig. 1) (9). In time-of-flight (TOF) PET acquisition, when two γ-photons are detected within a narrow coincidence time window, typically in nanoseconds (12-14), the difference in their arrival times provides the 1D spatial coordinate of the emission point along the LOR, as illustrated in Fig. 2.1A (1, 15). Locating separate annihilation events along their respective LORs can significantly increase the signal-to-noise ratio and reduce unwanted noise correlation during image reconstruction.

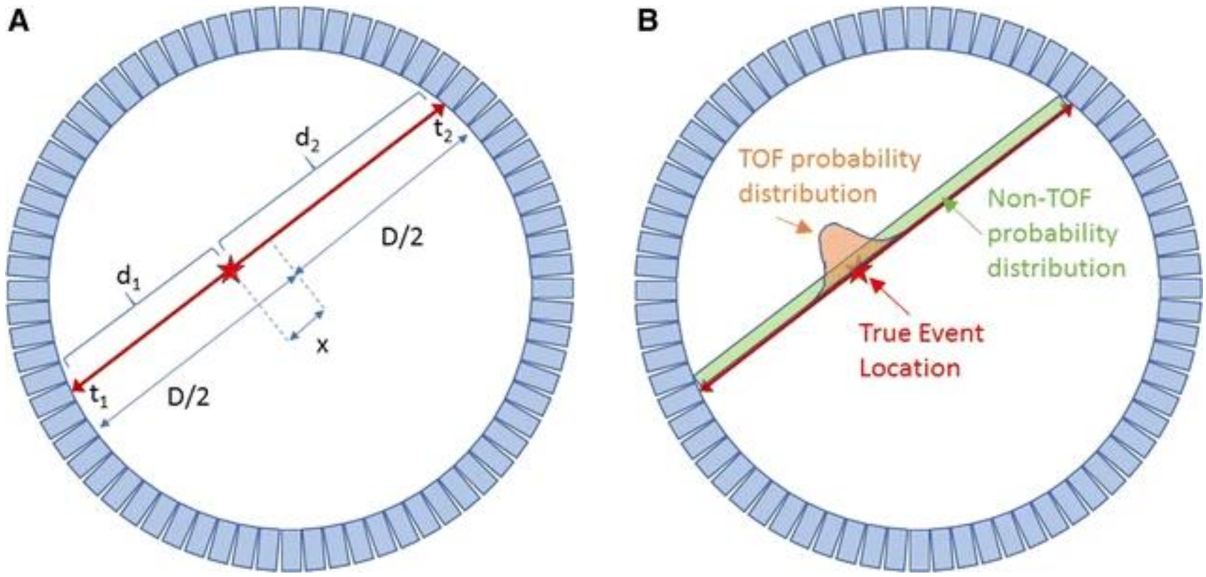

Fig. 2.1 Annihilation and TOF-PET acquisition (15).

Fig. 2.1A. Two photons emitted from the annihilation point (marked by a star) have respective arrival times of t_1 and t_2. The difference between t_1 and t_2 is used to estimate the emission location along the LOR using $x = c \times \frac{t_2 - t_1}{2}$, where c is the speed of light and x is the displacement of the annihilation point from the centre of the LOR.

Fig. 2.1B. The temporal resolution of non-TOF PET is limited, so it cannot measure the difference in arrival times between the two γ-photons. This results in an evenly distributed probability (green) along the LOR during image reconstruction. In contrast, TOF-PET assigns photon counts to a Gaussian distribution (red) centred around the true annihilation point.

2.2 Detection of Photons and Image Distortion

The detection of γ-photons relies on the photoelectric and Compton effects (8). Each detector contains a scintillator, and when a γ-photon reaches it, the photon deposits energy into the electrons of the scintillator crystal. The excited electrons transition to a higher energy state and then decay after a characteristic time, during which lower-energy photons are emitted, making the scintillator luminesce. These emitted photons reach the surface of a photodetector, generating an electrical signal that records the deposited energy and detection timing.

During reconstruction, the detected coincidences can be organised into sinograms or list-mode data (16). LORs can be represented by a point in a polar coordinate system and binned into sinogram bins. As shown in Fig. 2.2, the distance from the LOR to the centre of the field of view (FOV) and the azimuthal angle between each LOR and the horizontal axis are recorded. When a large number of data points from a single point source are recorded, they form a sinusoidal wave on the sinogram. List-mode, however, does not bin the detected coincidence events into sinograms but records a list of each detected event and the raw attributes of each coincidence instead (16). Providing motion correction and TOF information, list-mode is widely used in dynamic or iterative reconstruction with high spatial and temporal resolution.

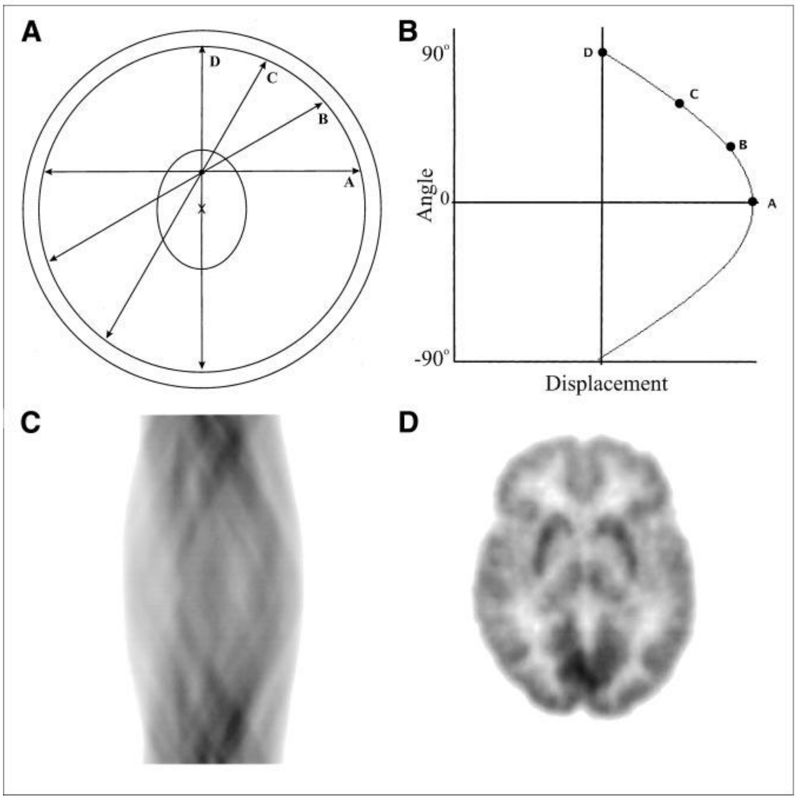

Fig. 2.2 Sinogram formation (10).

Fig. 2.2A. Four LORs originate from an annihilation point within the target volume.

Fig. 2.2B. The perpendicular distances from lines A, B, C, and D to the centre of the FOV (marked by a cross) are recorded as the x-coordinates of points A, B, C, and D in Fig. 2.2B. The azimuthal angle of each LOR is recorded as the y-coordinate of each point. Notably, the x-coordinate of point D equals zero where line D intersects the centre of the FOV; the y-coordinate of point A equals zero, meaning line A in Fig. 2.2A is horizontal.

Fig. 2.2C. A sinogram with overlapping sinusoidal waves generated from many emission points.

Fig. 2.2D. An illustration of a brain image reconstructed from the sinogram in Fig. 2.2C.

However, before γ-photons reach the detectors, they travel through the body and undergo attenuation and Compton scattering. Photons emitted deeper within the body experience greater attenuation by soft tissue than those emitted closer to the surface. The variable amount of attenuation as photons travel through tissues can be corrected by an integrated computed tomography (CT) system, where a transmission scan with X-ray CT provides a map of linear attenuation coefficients to correct photon attenuation (6-8, 12).

In contrast to attenuation, Compton scattering presents greater challenges for successful imaging. Compton scattering occurs when a γ-photon interacts with an outer-shell electron, transfers part of its energy, and gets deflected (5, 17). If the scattered photon still has enough energy, this process may repeat. Fig. 2.3 demonstrates that Compton scattering results in incorrect assignment of a LOR for the coincidence event and misplaced detection information (1). This is a major cause of noise, image artefacts, and reduced resolution. Without adequate scatter correction, scatter events can increase the risk of misdiagnosis, worsen lesion detectability, and increase uncertainty in cancer monitoring. A reliable protocol that effectively corrects for scatter is therefore essential for high-precision disease diagnosis and prognosis with PET.

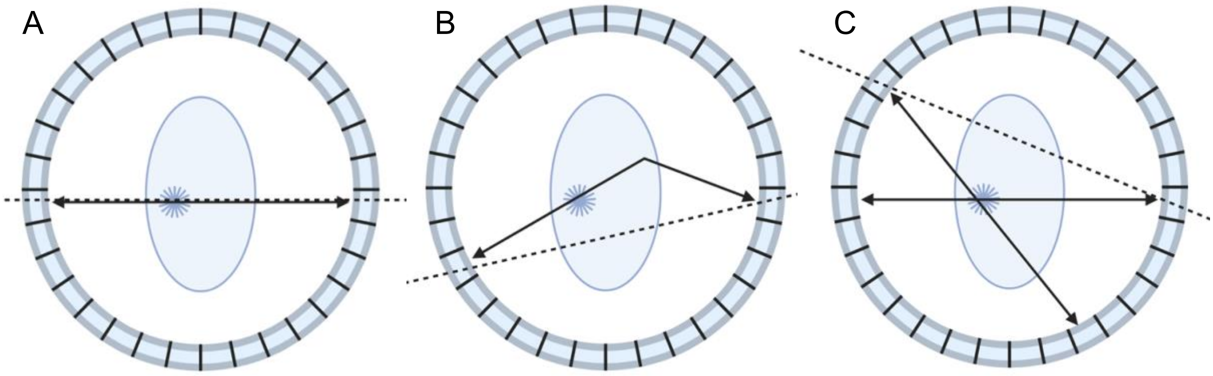

Fig. 2.3 The coincidence and scatter events of PET (created with BioRender.com). The images illustrate three possible scenarios of coincidence events within a detector ring.

Fig. 2.3A. True coincidence: two γ-photons from an annihilation point travel inside the plane of the detector ring in the direction of the arrows without scattering, reach the detector ring, and define a LOR (the dashed line).

Fig. 2.3B. Scatter coincidence: one of the two γ-photons scatters and gets deflected, resulting in an incorrect LOR (the dashed line).

Fig. 2.3C. Random coincidence: two photons from different annihilation events reach the detector ring within the coincidence time window, and an incorrect LOR is registered.

The concerns of scattering are amplified in fully 3D PET (Supplementary Fig. 2). In 3D PET, coincidence events along all possible LORs within the target volume are simultaneously measured during image acquisition, unlike the slice-by-slice reconstruction approach in septa-containing, slice-collimated PET (Supplementary Fig. 3) (8, 11, 18-21). Fully 3D PET has significantly addressed the demand for higher sensitivity, with the overall number of detected coincidence events increasing by 5- to 10-fold compared with 2D PET (11, 22-24). However, this gain in sensitivity comes at the cost of a substantial increase in the fraction of scatter events, rising from 10–20% in 2D PET typically increases to 40–60% in 3D PET (23, 25). This increased scatter fraction further necessitates reliable scatter correction for the widespread application of PET as a clinical diagnostic tool.

2.3 Scatter Correction

2.3.1 Single Scatter Simulation

One of the most widely used approaches for scatter correction is single scatter simulation (SSS) (5, 17-20, 26). SSS assumes that only one of the two photons undergoes Compton scattering only once before detection, where such single-scatter events are considered the primary contributors to detectable scatter. SSS begins by reconstructing the emission data with attenuation correction but without scatter correction. The algorithm then simulates single-scatter events and generates a scatter estimate based on the attenuation map and the non-corrected image.

To compensate for multiple-scatter events, SSS requires sinogram tail-fitting. The tail region of the estimated sinogram, where the detected signal is predominantly scattered, is scaled to match the measured data, and this scale factor is applied to the entire SSS estimate. A scatter-corrected image is then generated by subtracting the scaled SSS estimate from the measured data, producing corrected sinograms for reconstruction. This process can be iterated until an image with reasonable scatter correction is obtained (5, 18, 27).

In SSS, the Klein–Nishina formula describes the probability of photon scattering along every LOR and forms a sinogram of scatter events (18, 19):

$$\frac{d\sigma}{d\Omega} = \frac{r_e^2}{2}\left(\frac{E^{'}}{E}\right)^2\left(\frac{E^{'}}{E} + \frac{E}{E^{'}} - \sin^2\theta\right) \quad (1)$$

This formula expresses that the fraction of the total number of scattered particles emerging in the solid angle $d\Omega$ depends on the photon energies before scattering (E) and after scattering (E'), the electron radius r_e, and the scatter angle θ.

Despite its accuracy and robustness, SSS exhibits several limitations in more complex imaging scenarios. SSS's inability to accurately model multiple-scatter events becomes especially concerning when imaging patients instead of tissue samples, especially in whole-body PET scans (17). The increased heterogeneity and volume of patients lead to a higher probability of multiple scattering, reducing the accuracy of tail-fitting (5, 17, 19). Additionally, if there is any patient motion between the transmission scan and the emission scan, the attenuation map that SSS relies on is no longer spatially aligned with the true distribution of the tracer (5, 20). SSS also cannot provide an accurate estimate when there is a sharp change in activity concentration, for example the boundary of a radio-labelled organ or a “hot” phantom (19). Finally, SSS is unable to model scattered photons that originate from outside of the axial FOV, and scatter estimation is therefore often underestimated (5, 20).

The inaccuracies of SSS can lead to significant PET image artefacts and serious quantification errors in reconstructed images, potentially masking tumours and local recurrences (28). Alternative approaches that attempt to resolve these limitations are being developed. More recent scatter correction approaches using deep learning have shown promising results (5, 20, 25), but deep-learning-based scatter estimation (DLSE) requires a large and realistic training dataset (19). Several studies have investigated energy-based scatter estimation (EBSE) to develop a scatter correction algorithm that addresses many of the concerns associated with SSS (5, 19, 20, 29-31).

2.3.2 Energy-based Scatter Estimation

Immediately after annihilation, each emitted photon has an energy of 511 keV. Unscattered photons reach the detector at this initial energy, but due to the energy resolution of the detector, the measured spectrum is modelled by a Gaussian function centred at 511 keV (19). In contrast, scattered photons lose part of their energy, resulting in a spectrum with more coincidence counts at lower energy levels, as shown in Fig. 2.4. In traditional non-energy-based scatter correction, energy information is only used to reject low-energy photons (18), but EBSE extracts the energy information of detected coincidence events to analyse scatter history. Early EBSE approaches were typically limited to a few energy windows (5, 29, 31, 32). Recent computational developments have boosted the investigation of EBSE by storing list-mode data, enabling the precise specification of energy windows and more effective use of detailed energy information (5, 20, 30). Some recent EBSE methods are parametric, using list-mode data and detailed energy records (5, 19, 20, 30). Our work evaluates a parametric approach and calculates the contributions of unscattered and scattered photons for more accurate scatter estimation (19).

Since each coincidence event involves two detectors, counts from all possible detector pairs are combined across all energy windows to produce 2D histograms (5, 20, 30). However, as described in this section, fitting probability density functions (PDF) in 2D histograms is computationally intensive and noisy, particularly in pixels with few counts. As a practical modelling alternative, one dimension of the 2D histogram is usually marginalised, providing greater robustness and practicality while preserving the characteristics of the energy spectra (19, 20), as shown in Fig. 2.4.

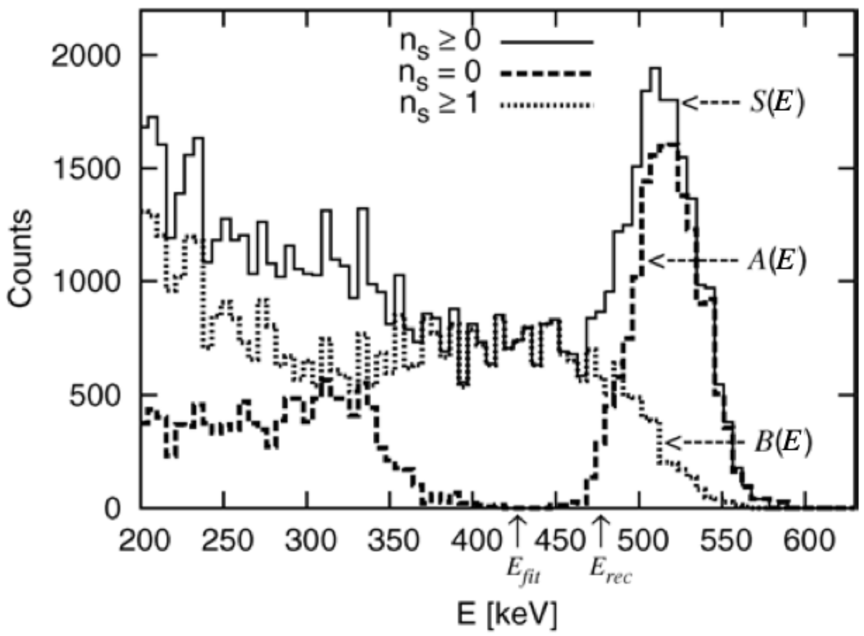

Fig. 2.4 The 1D energy spectra of one detected photon (32).

Our study focuses on the energy information linked to the number of scatters (n_s) of each photon. The spectrum of unscattered photons is represented by $A(E)$ in the dashed line ($n_s = 0$), where a peak is present at around 511 keV. $B(E)$ in the dotted line represents the number of photons that undergo at least one scatter ($n_s \geq 1$), where there are more coincidence counts at lower energy levels. These two spectra are combined, with respective coefficients, into a spectrum that contains all detected prompt events represented by the solid line $S(E)$.

E_{fit} marks a certain energy threshold above which the variation of $B(E)$ is limited. E_{rec} is the lower limit of the photopeak; only photons above this boundary are used for image reconstruction.

We specifically focused on modelling the joint energy histogram of all events as a linear combination of four scenarios proposed by Efthimiou et al. (19). Under the assumption that the two emitted photons are mutually uncorrelated, the events can be classified into four scenarios:

1. Both photons are unscattered (non-scatter events);
2. Photon 1 scattered, photon 2 unscattered (single-scatter events);
3. Photon 1 unscattered, photon 2 scattered (single-scatter events);
4. Both photons are scattered (double-scatter events).

Each scenario is weighted by respective coefficients estimated through statistical fitting (19).

EBSE does not require an attenuation map, so it successfully improves the joint estimation and analysis of activity and attenuation, providing a resolution to many misdiagnosis-related concerns in SSS (5, 33). EBSE is especially advantageous in circumstances without high-resolution CT scanners or where additional exposure to X-rays is undesirable. Notably, since the CT-generated attenuation map is integrated into scatter estimation, EBSE can serve as a suitable alternative to SSS.

2.4 Simulation Platforms

In this work, we simulated simplified 3D PET images using the Docker image of Synergistic Image Reconstruction Framework (SIRF). SIRF is an open-source library designed to unify access to various tomographic image reconstruction engines across multiple imaging modalities, especially PET and magnetic resonance (MR) in synergy (2). SIRF interfaces in Python with the reconstruction engine for PET applications, Software for Tomographic Image Reconstruction (STIR), to perform physics modelling and handle data input and output (34).

STIR contains a range of C++ classes for PET reconstruction, including scanner models, attenuation correction, scatter modelling, and synthetic phantom simulation. STIR can generate a 3D voxelised phantom image, support file conversion and manipulation, perform SSS-based scatter correction, and export image data for further analysis. This setup specifies the distributions and dimensions of activity and attenuation in a fully controlled and reproducible environment (35). Controlled virtual experiments with explicitly defined image geometry, tracer distribution, and physical effects facilitate the prototyping and validation of various image reconstruction algorithms. SIRF and STIR therefore provide an essential foundation for visualising and validating correction techniques, testing reconstruction inputs, and developing preprocessing pipelines in the absence of real patient or scanner data. In this study, the ECAT HR+ scanner (also known as ECAT 962), developed by CTI-Siemens, was used in all simulations. ECAT HR+ provides excellent and isotropic spatial resolution, making it well suited for evaluating scatter-correction algorithms in 3D PET and whole-body imaging (36, 37).

While STIR is capable of simulating PET emission data, it is limited to simplified models and single-scatter events. A Monte Carlo physics-based simulator called Simulation System for Emission Tomography (SimSET), developed by University of Washington (38), was used to generate more realistic and complicated simulations. Full Monte Carlo simulation is an important tool that is gaining increasing attention in PET imaging (13, 19, 20, 30-32, 39, 40). Instead of producing only sinograms, Monte Carlo simulation generates detailed photon histories in list-mode in order to maximise spatial and temporal resolution (39). The capability for highly realistic modelling of Compton scattering makes Monte Carlo simulation critical for advanced reconstruction and scatter correction. In this project, SimSET outputs were converted to sinograms to evaluate scatter correction methods under fully controlled and realistic conditions without relying on clinical scanners or patient data. The simulated ground truth of emission and scatter distributions was then compared with our estimates.

However, a key drawback of full Monte Carlo simulation is that accurate physical modelling is extremely computationally demanding, making a complete simulation cycle very time-consuming (19). Among the hundreds of millions of simulated photons, only a small fraction result in detection events. The vast majority are either absorbed in tissue or scattered outside the detection range, creating large variance between simulations and computational waste (41). To address this problem, importance sampling techniques are applied in SimSET, including stratification, forced detection, and forced non-absorption. Importance sampling techniques redirect resources towards photon histories that contribute to successful detection events, such as those within the scanner's acceptance angle. Importance sampling has been reported to improve computational efficiency by 10- to 300-fold and to minimise wasted simulation of irrelevant histories (39, 40).

The SimSET software consists of multiple modular components, as shown in Fig. 2.5 (38). The Photon History Generator (PHG) of SimSET incorporates the attenuation image into a Monte Carlo framework. Using the spatial information provided by activity and attenuation images, PHG virtually generates one β^+ decay and two γ-photons at a time and simulates the stochastic photon interactions and the travel of the photons through heterogeneous attenuators (38).

The Collimator module tracks photon absorption and penetration as they pass through a type of collimator, for example cylindrical septa or endplates, using a set of separate Monte Carlo techniques. In our simulations, the Collimator module was not activated. The Detector module specifies the detector type, receives photons from PHG or Collimator, and tracks how photons interact with the detectors (38). Next, the Binning module aggregates photon histories into various sinograms by assigning a weight to each decay event at each specific energy level. The weight data indicate how many decays in a real-world scanner each simulated decay represents based on activity and simulation duration as a fraction. The weight as well as the number of counts in each bin are stored because of the down-sampling in PHG simulation and the presence of importance-sampling biases (38). The sum of squared weights is also recorded to provide a more accurate estimation of weight variance.

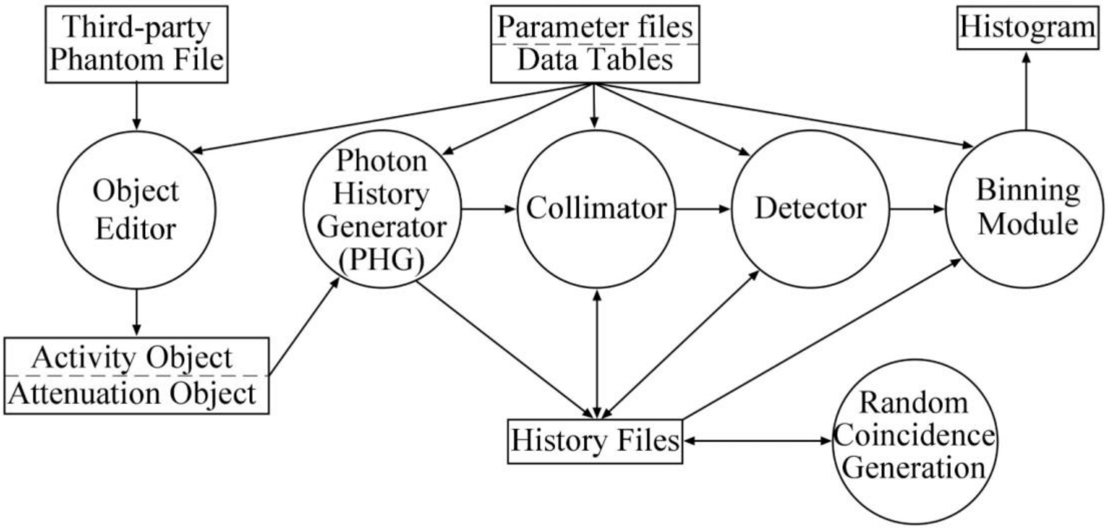

Fig. 2.5 The workflow of SimSET (38).

The Photon History Generator is the central module of SimSET, which takes the activity and attenuation images from the Object Editor module and models the generation and transport of photons to the Collimator and Detector modules (40, 41).

All modules are controlled by their respective parameter files. PHG, Collimator, and Detector can produce photon history files containing decays and photons in list-mode if a simulation is paused; these files can then be used as input to resume the simulation. The Random Coincidence Generation module can also simulate random coincidences between the Detector and Binning modules.

2.5 The Method of Moments

The method of moments (MoM) is a statistical parameter estimation technique. It equates the sample moments from the measured data to theoretical population moments and solves for distribution parameters. Moments describe the shape of PDF, where the first raw moment is the mean of the distribution, given by $\mu_n' = E[X^n]$. The second-order central moment characterises the variance of the distribution, whereas higher-order standardised moments are related to skewness and kurtosis (42).

In the context of Efthimiou's work, MoM was used to avoid the complication of fitting 2D PDF (19). Instead, moments were calculated from PDF at discrete energies and equated to the moments of the joint energy distribution $C(E_1, E_2)$. The solution of this matrix equation was then used to generate scatter sinograms for each sinogram bin. This is a multivariate extension of moments, where MoM matches the joint raw moments of E_1 and E_2 between the ground truth and the model estimate. the coefficients of all scatter events along individual LORs or small groups of LORs were aggregated into a scatter sinogram. In the reconstructed image, the scatter sinogram generated from the scatter PDF would be subtracted to yield a scatter-corrected image.

Our work evaluated the method of Efthimiou et al. in a 1D energy-based approach. Under the assumption that PDF of the two photons are independent, the non-scatter PDF $U(E)$ was fitted with a Gaussian function, while the data-dependent scatter PDF $S(E)$ in 1D was derived from 1D histograms with sufficient statistical robustness (19).

Once 1D $U(E)$ and $S(E)$ were estimated, they were combined into a 2D representation, where MoM captured the multiplicative effects of E_1 and E_2 on the joint energy distribution of all photon pairs.

3. Methods

3.1 Environment Setup

Miniconda was downloaded with the following commands (43).

```
mkdir -p ~/miniconda3

curl https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-arm64.sh

-o ~/miniconda3/miniconda.sh

bash ~/miniconda3/miniconda.sh -b -u -p ~/miniconda3

rm ~/miniconda3/miniconda.sh
```

A conda environment named `stirenv` was configured, within which STIR version 6.1.0 and plotting utilities were installed from `conda-forge`.

`conda create --name stirenv -c conda-forge stir matplotlib` (44)

The simulations and reconstructions were performed inside the Docker container `sirf:latest` (version 3.7.0). A Jupyter web interface (host port 9999 mapped to container port 8888) was used to access the container's Python environment.

`docker run --rm -it -p 9999:8888 ghcr.io/synerbi/sirf:latest` (45)

SimSET version 2.9.2 was downloaded and uncompressed from the SimSET official website:

```
anymachine% uncompress phg.2.9.2.tar.Z

anymachine% tar -xvf phg.2.9.2.tar
```

Within the uncompressed folder `2.9.2` (see the GitHub link in Appendix), the subfolder called `make.files` contains `simset.make`, which controls the SimSET directory:

```
SIMSET_PATH = /Users/useruser/Desktop/2.9.2
```

The directory should be replaced with a specified installation path.

`simset.make` also defines an operating-system-specific compiler flag needed for building SimSET. For macOS Darwin that SimSET was currently set to run on, the compiler flag is:

```
OS_CFLAGS = -DDARWIN -g
```

After defining the directory and compiler, navigate to folder `2.9.2` and run the script `make_all.sh` to build SimSET. For demonstration, the default directory is used here:

```
cd /Users/useruser/Desktop/2.9.2
./make_all.sh
```

If the compilation does not report any errors, run `runFast.sh` to validate the installation with the sample data provided:

```
cd ./samples/fastTest
./runFast.sh
```

This validation can run on little-endian CPUs. The results of validation are contained in the log files of the nine simulations initiated by `runFast.sh`.

3.2 Generate Attenuation and Activity Images

Our validation began with a narrow line source embedded in an identical narrow attenuation cylinder. This configuration minimises scatter to robustly fit the distribution of unscattered photon energy, excluding any significant effects of scattered photons. With appropriate cylinder radii, this setup provides scatter-free data for reconstruction, which can be used as a benchmark for EBSE and other scatter correction algorithms.

In the main simulations, a larger attenuation cylinder was introduced to emulate realistic scattering scenarios and quantify the contributions of unscattered and scattered photons to the joint energy spectrum. STIR-generated attenuation coefficients in cm^{-1} were converted to a SimSET-compatible, pre-defined material index, including water, bone, air, and aluminium (35, 38).

Within the Docker image, the module `sirf.STIR` was imported to initiate reconstruction and data manipulation. The conventional of STIR data file format `.s` contains sinogram data, and each data file has a corresponding human-readable interfile header `.hs`, which records variables such as the scanner geometry, detector dimensions, and the index of the energy window. The scanner type was set to ECAT 962. Some scanner parameters are listed as follows:

Table 3.1 The scanner parameter used to simulate attenuation and activity images

Scanner Geometry	Cylindrical
Number of rings	12
Number of detectors per ring	64
Inner ring diameter (cm)	81.02
Distance between rings (cm)	1.308
Energy resolution	0.1
Reference energy (keV)	511
Energy window lower level (keV)	400
Energy window upper level (keV)	600

The energy resolution determines how well a detector can distinguish photons with different energies. It was quantified by the width of the photopeak relative to the reference energy, which is discussed in more detail in Section 3.5. The class `AcquisitionData` read the header file `scanner.hs`, loaded the sinogram dataset, and generated a compatible acquisition template that matched the scanner geometry. Based on this template, an empty 3D image grid with inherited dimensions and geometry was created by `ImageData`. A volume of attenuation material, in this case water, was added to this image via `add_shape` to describe the attenuation distribution.

Our images were generated from the acquisition template with 379 pixels in both the x- and y-directions and 23 pixels in the z-direction. The numbers of voxels in the z-, y-, and x-directions of the image were sequentially obtained and stored in the variable `image_size` as a three-element tuple. The physical dimensions of each voxel were 6.54 mm, 2.13 mm, and 2.13 mm in the z-, y-, and x-directions and were stored as a tuple in the variable `voxel_size`. The `image_size` and `voxel_size` variables were collectively used to specify and scale the radii of elliptical cylinders in the activity and attenuation images. The activity image was cloned from the attenuation image, which described the spatial distribution of radiotracers within the cylinder. This process ensured that the activity and attenuation images had identical dimensions specified in 3D Cartesian coordinates for each round of simulation with SimSET.

We started with a trial simulation that generated a narrow cylinder with a radius of 8.07 mm in both the activity and attenuation images, a line source in air with little surrounding scatter, as shown in Fig. 3.1A and 3.1B. By specifying the z-value as the centre point of the axial FOV, the images were displayed as an x–y cross-section at the midpoint of the z-axis. This configuration represents the scenario where the spatial distribution of the radiotracer is uniform within the emission source, with little or no surrounding attenuation from water. This trial simulation was used to validate that the simulation environment had been set up successfully.

The images were converted to arrays and visualised using `show_2D_array`, a utility of SIRF (2). The attenuation image contained information about the attenuation medium, in this case water, which was scaled by a coefficient of 0.09687 cm^{-1} to model photons travelling through water. A cylindrical water phantom was created and added to the attenuation image. In Fig. 3.1A, the attenuation image contains the linear attenuation coefficients measured in cm^{-1}, where the attenuation at the yellow dot has a coefficient of about 0.097 cm^{-1}; the activity image displays the rate of radioactive decay in becquerels per cubic centimetre ($Bq{\cdot}cm^{-3}$).

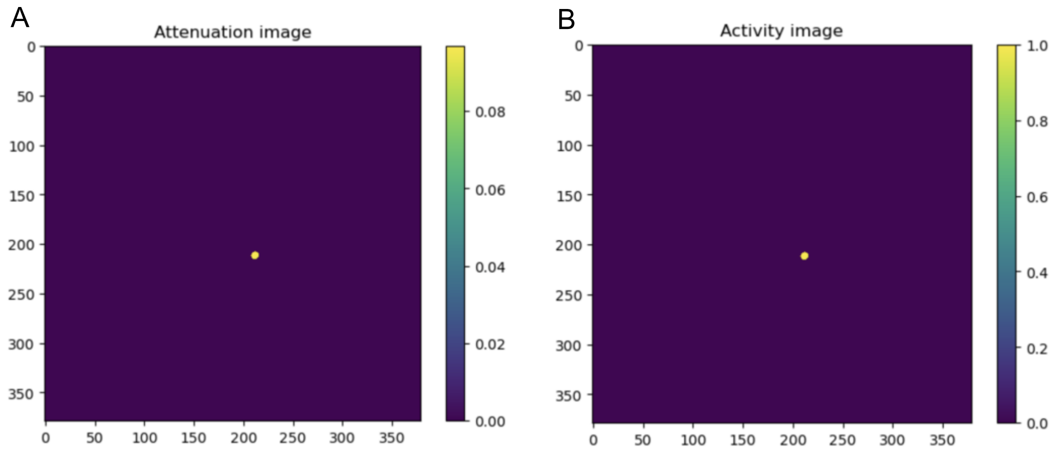

Fig. 3.1 The activity and attenuation images with minimal scattering.

Fig. 3.1A. The attenuation image contains a narrow cylinder of attenuation medium.

Fig. 3.1B. A yellow dot in the activity image marks the location of the emission source.

During the main simulation, the radius of the attenuation cylinder was increased from 8.07 mm to 150 mm, while the radius of the activity cylinder was increased to 24.21 mm. The increased attenuation medium led to a higher number of scatter events, allowing us to better quantify the contribution of scattered photons to the total spectrum. The length of both cylinders remained at 150.42 mm. Centred at the origin where $z =$

0, the cylinder spanned from +75.21 mm to –75.21 mm along the z-axis. Only the half of the cylinder located on the positive side of the z-axis was stored and displayed within the axial FOV. The images were converted to NumPy arrays and displayed in Fig. 3.2A and 3.2B. The activity and attenuation images were then exported for SSS and SimSET simulations.

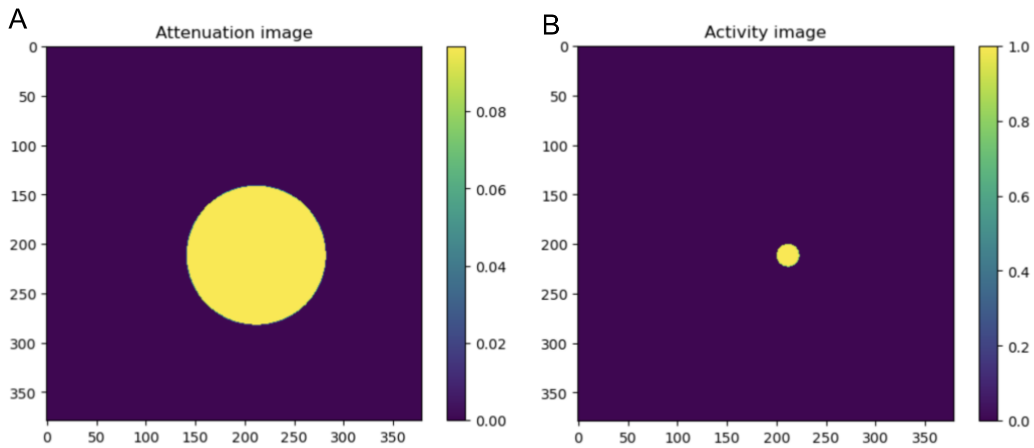

Fig. 3.2 The attenuation and activity images used in the main simulation

Fig. 3.2A. An attenuation cylinder with a radius of 150 mm was simulated. The attenuation coefficients within this cylinder are uniformly 0.096 cm^{-1}.

Fig. 3.2B. The activity image shows the uniform distribution of radioactive decay within the cylinder at 1.0 $Bq{\cdot}cm^{-3}$.

Although the radii were increased, the position of the cylinder origins remained unchanged.

3.3 Single Scatter Simulation via STIR

Based on the simulated images shown in Fig. 3.2A and 3.2B, the STIR class `SingleScatterSimulator` was used to perform scatter simulation. An attenuation map was constructed from the attenuation image, and the acquisition template and activity images were input into the single scatter simulation.

`SingleScatterSimulator` outputs slices of scatter sinograms represented by a 4D array indexed as `[TOF-bin, sinogram, view, tangential]` (35). One slice was viewed using the index `[0,5,:,:]`. Here, index 5 selected the sinogram corresponding to the sixth slice located at the midpoint of all twelve slices. This index displayed a cross-section with `view` and `tangential` as the two dimensions. The number of views of a ring (the number of azimuthal bins) is half the number of detectors on each ring due to interleaving (35), as specified in Table 3.1. In the sinogram, `view` is the y-axis, and `tangential` refers to the bin within a view along the horizontal direction.

3.4 SimSET Simulations

Positron-emitting ^{18}F was used in all simulations. For each round of simulation, the attenuation image was cloned to prevent overwriting, and all simulations were performed only on the cloned image. Previous outputs and the cloned attenuation image were erased before each run. A new folder was created and specified as the output directory. The STIR utility `generate_image` read the parameter file and generated a uniform cylinder with the following dimensions:

Table 3.2 The geometry of SimSET-simulated uniform cylinder

Shape	Ellipsoidal cylinder
Dimensions (x-, y-, and z-axes)	111 × 111 × 47 voxels
Voxel size (x-, y-, and z-axes in mm)	3 × 3 × 3.27
Radius-x (mm)	100
Radius-y (mm)	100
Origin (mm)	(70, 10, 20)
Axial length (mm)	110

The utility `stir_math` with the option `--times-scalar 0.096` was used to uniformly scale the cylinder based on the attenuation coefficient of water, creating `my_uniform_cylinder.hv`. Once the integrity of the input data had been verified, the simulation proceeded. The activity image was converted to byte files, while the cloned attenuation image was converted to SimSET indices using `conv_to_SimSET_att_image`. Within the template parameter file `template_phg.rec`, variables such as input and output directories, the number of simulated photons, and parameter file names were replaced with the actual inputs for this round of simulation. These variables were then written into a new ASCII text file, `phg.rec`, which was the parameter file read by PHG.

With importance sampling enabled, the core PHG executable of SimSET read `phg.rec` and performed the simulation. Some of the simulation parameters are listed as follows:

Table 3.3 The parameters of SimSET

Z-axis bins	12
Energy bins	10
Azimuthal angle bins	32
Transaxial bins	55
target_zMin (cm)	-0.654
target_zMax (cm)	15.042
Radius (cm)	40.51
Acceptance angle	90°
Detector type	1
scatter_parameter	3
min_s	0
max_s	2

The minimum and maximum edges of the target cylinder on the z-axis (`target_zMin` and `target_zMax`) were determined by the ring spacing and number of rings (the number of z-axis bins). This configuration ensured that each output file consisted of

12 sinograms. The Binning module fixed `target_zMin` at $-\frac{1}{2} \times (\text{ring spacing})$, and from that point twelve scanner rings were placed with a spacing of 1.308 cm, as specified in Table 3.1. Geometry details such as edge coordinates and voxel numbers were automatically converted to the PHG input format by the STIR utilities `list_mode_info` and `write_phg_image_info`. The number of azimuthal angle bins was half of the total number of detectors per ring. The number of transaxial bins referred to the number of bins along each LOR. The acceptance angle was set to 90° to maximise LOR detection and sensitivity. `Detector type = 1` applied a simple Gaussian blur to the incident energy (38).

In the Binning module of SimSET, setting `scatter_param = 3` enabled better specification of scatter events. Photons with a scatter number greater than `max_s` were binned together. Specifically, in the photon history file, one emitted photon was arbitrarily labelled “blue” and the other “pink” (the SimSET convention). This mode distinguished between the two types of single-scatter events and separated double-scatter events from single-scatter events. Four scenarios were specified, where the scattered photon was labelled as “1” in the file names:

1. non-scatter events: `blue0_pink0.hs`;
2. single-scatter events: `blue0_pink1.hs` (the “pink” photon scattered);
3. single-scatter events: `blue1_pink0.hs` (the “blue” photon scattered);
4. double-scatter events: `blue1_pink1.hs`.

These output files (`.hs` and `.s`) were uploaded into a directory

`/home/jovyan/work/8.10/scatter_param=3` within the Jupyter interface, as described in Section 3.1.

Specific output files were then extracted and converted back to STIR format for further analysis. `weight`, `activity_image`, and `attenuation_image` were extracted from the output directories. The weight files were input into the shell script `conv_SimSET_projdata_to_STIR_modified.sh`, which read the scatter setup and wrote the weight data into multiple STIR-compatible sinogram files. For each of the four scenarios, weight values were binned into a 10×10 array of files, with one sinogram for each of the ten energy windows on either detector. The script `make_hv_from_SimSET_param.sh` read the directories of `activity_image` and `attenuation_image`, extracted the output files of SimSET, and converted them into a STIR-compatible format. Geometry information contained in the SimSET output was preserved and interpreted by STIR.

The detected photons were binned into ten energy windows with a width of 20 keV, spanning from 400 to 600 keV at each detector, forming 2D histograms. Specifically, four 10×10 NumPy arrays were initialised with zeros as the template histograms for `blue0_pink0`, `blue0_pink1`, `blue1_pink0`, and `blue1_pink1`. The STIR class `AcquisitionData` of STIR extracted the acquired simulation data from the corresponding files as raw sinograms. In the first step, for each of these four arrays, a 2D histogram was constructed by combining all LORs in all sinogram bins within that energy window using the `sum()` function. The summed value replaced zero at corresponding coordinates in the 2D histogram. `blue0_pink0` represented the non-scatter component of all detected coincidences and was stored as `nonscatter_histogram`; the scatter component, `scatter_histogram`, was the sum of the other three arrays. `nonscatter_histogram` and `scatter_histogram` were then combined into `total_histogram`, which stored the summed weights of all detected photons. These three histograms were visualised with `matplotlib.imshow()`.

The 2D histograms were then reduced into 1D arrays. The non-scatter and scatter components were combined to generate a third array. Seven arrays–`nonscatter_histogram`, `scatter_histogram`, `total_histogram`, `blue0_pink0`, `blue0_pink1`, `blue1_pink0`, and `blue1_pink1`–were reduced to 1D using `numpy.sum(..., axis=1)`. The first three 1D arrays were visualised as spectra representing the weights across the ten energy windows of E_1.

3.5 Estimating Non-scatter and Scatter PDF

The joint energy histogram is given as follows (19):

$$C(E_1,E_2) = c_0U(E_1)U(E_2) + c_1U(E_1)S(E_2) + c_2S(E_1)U(E_2) + c_3S(E_1)S(E_2) \quad (2)$$

In this equation, $U(E)$ and $S(E)$ are the PDF of unscattered and scattered photons, respectively. The PDF represents the probability that a detected photon of a specific energy is scattered or unscattered. c_0 is the weight of non-scatter events, c_1 and c_2 represent the weight of single-scatter events, and double-scatter events have weight c_3. Two γ-photons are detected with energies E_1 and E_2. The PDF and the photon counts are combined to quantify the contributions of scattered and unscattered photons to the total energy spectrum. Random coincidence events were not simulated in our investigation.

The `det.rec` parameter file set the Detector module to model a Gaussian distribution, as specified in Table 3.1. The energy resolution at 511 keV was 10%, meaning that the full width at half maximum (FWHM) was given by:

$$\mathrm{FWHM} = 10\% \times 511\ \mathrm{keV} = 51.1\ \mathrm{keV} \quad (3)$$

This reflects the detector's capability to distinguish between photons with similar energies. The FWHM was used in the calculation of the standard deviation of energy resolution at 511 keV (σ_{511}) (19, 46):

$$\sigma_{511} = \frac{\mathrm{FWHM}}{\sqrt{8 \ln 2}} \approx 21.7\ \mathrm{keV} \quad (4)$$

Across ten energy windows, the estimated non-scatter PDF $U(E)$ is given by:

$$\mathrm{U}(E) = \frac{1}{\sqrt{2\pi} \cdot \sigma_{511}} \cdot e^{-\frac{(E_i - 511)^2}{2\sigma_{511}^2}} \quad (5)$$

where E_i was the median value of each energy window (410, 430, 450 keV, etc.), and these values were stored in a list called `energies`. The non-scatter PDF was derived from a Gaussian function and was already normalised (area under the curve ≈ 1), verified via numerical integration with `np.trapz(PDF_nonscatter, energies)`.

By summing over E_2, equation 2 was re-written as:

$$\mathrm{C}(E_1) = (\mathrm{c}_0 + \mathrm{c}_1)\mathrm{U}(E_1) + (c_2 + c_3)\mathrm{S}(E_1)$$

or equivalently:

$$\mathrm{C}(\mathrm{E}_1) = \mathrm{c}_0\mathrm{U}(\mathrm{E}_1) + \mathrm{c}_1\mathrm{U}(\mathrm{E}_1) + c_2\mathrm{S}(E_1) + c_3\mathrm{S}(E_1) \quad (6)$$

The 1D arrays of the ground truth corresponded to the four terms in equation 6:

$$blue0\_pink0\_1D = c_0 U(E_1)$$

$$blue0\_pink1\_1D = c_1 U(E_1)$$

$$blue1\_pink0\_1D = c_2 S(E_1)$$

$$blue1\_pink1\_1D = c_3 S(E_1)$$

In 1D, the scatter PDF of E_2 was marginalised because of the normalisation, so `blue0_pink0` and `blue0_pink1` were combined to fit the non-scatter component of the total spectrum. This was normalised with `np.trapz(blue0_pink0_1D + blue0_pink1_1D, energies)` and plotted together with $U(E)$. The shapes of the two curves were compared to assess how well the Gaussian model in equation 5 matched the normalised non-scatter component in the ground truth.

To estimate $S(E)$, the non-scatter component was subtracted from the total histogram. We compared the three 1D curves and selected an energy range where the total spectrum was primarily unscattered photons and scattered photons at very small scatter angles. In that range, the scale factor c_0' approximated the total contribution of unscattered photons in the ground truth $(c_0 + c_1)$. Combining Efthimiou's work (19) and the 1D curves, we selected only $+4\sigma_{511}$ of the photopeak as our energy window for the range of unscattered photons, which corresponded to windows 6 to 9 (520-600 keV).

The values of c_0 and c_1 were calculated as:

$$c_0 = \frac{\sum_{i=6}^{9} \text{blue0\_pink0}}{\sum_{i=6}^{9} \text{PDF\_nonscatter}} \quad (7)$$

$$c_1 = \frac{\sum_{i=6}^{9} \text{blue0\_pink1}}{\sum_{i=6}^{9} \text{PDF\_nonscatter}} \quad (8)$$

Since most photons in windows 6–9 were unscattered, the total contribution of unscattered photons c_0' was calculated by directly scaling the non-scatter PDF to `total_1D_histogram`:

$$c_0' = \frac{\sum_{i=6}^{9} \text{total\_1D\_histogram}}{\sum_{i=6}^{9} \text{PDF\_nonscatter}} \quad (9)$$

We compared c_0' with $(c_0 + c_1)$ to validate the model.

The spectrum of unscattered photons was then subtracted from the total spectrum to estimate the spectrum of scattered photons:

$$(c_2 + c_3)S(E_1) = C(E_1) - c_0' U(E_1) \quad (10)$$

The values of $S(E)$ in a list were then normalised so that the area under the curve was equal to 1, producing the scatter PDF $S(E)$:

```
PDF_scatter_scaled = total_1D_histogram - np.array(PDF_nonscatter) *
unscatter_total_con
area = np.trapz(PDF_scatter_scaled, energies)
```

```
PDF_scatter = PDF_scatter_scaled / area
```

For validation, the 1D arrays of `blue1_pink0` and `blue1_pink1` were combined and normalised as the scatter component of total detected coincidence events. The estimated scatter PDF was then compared with the scatter component of the ground truth.

3.6 The Estimation of Scatter Sinograms with MoM

To estimate scattering in each LOR, MoM was applied in a 2D approach by equating population moments q_{m_1,m_2} (the model estimate) to the sample moment $\hat{q}_{m_1,m_2}$ (derived from the measured data). The sample moment $\hat{q}_{m_1,m_2}$ of each LOR with order (m_1, m_2) (20) was calculated as:

$$\hat{q}_{m_1,m_2} = \sum_{E_1}\sum_{E_2} E_1^{m_1} E_2^{m_2} C(E_1, E_2) \quad (11)$$

Where the measured energies E_1 and E_2 of two photons were the midpoint of each energy window, and their respective orders were m_1 and m_2. Instead of summing across all sinograms in each file for the 2D histograms, `AcquisitionData` extracted the weight recorded in each sinogram bin. The contribution of photon energy was determined by the moment orders:

1. When $(m_1, m_2) = (0,0)$, the energy of neither photon applies any weight, so $\hat{q}_{0,0}$ is simply the total weight in `total_histogram`.
2. For $(m_1, m_2) = (1,0)$ or $(m_1, m_2) = (0,1)$, $\hat{q}_{m_1,m_2}$ is scaled by E_1 or E_2. For example, if $(m_1, m_2) = (1,0)$, the total weight within each E_1 bin (each row of the

histogram) is multiplied by the midpoint of the respective bin. The sum of all scaled row weights gives $\hat{q}_{1,0}$.

3. If $(m_1, m_2) = (1,1)$, the weight in each pixel is scaled by E_1 and E_2 multiplicatively; summing across all pixels gives $\hat{q}_{1,1}$.

The theoretical moment q_{m_1,m_2} of the model in equation 2 (19) was calculated by:

$$\mathrm{q}_{\mathrm{m}_1,\mathrm{m}_2} = \int_\epsilon \int_\epsilon \mathrm{E}_1^{\mathrm{m}_1} \mathrm{E}_2^{\mathrm{m}_2} \, \mathrm{C}(\mathrm{E}_1, \mathrm{E}_2) \, \mathrm{dE}_1 \mathrm{dE}_2$$

$$= \sigma_0 a_{m_1} a_{m_2} + \sigma_1 a_{m_1} b_{m_2} + \sigma_2 b_{m_1} a_{m_2} + \sigma_3 b_{m_1} b_{m_2} \quad (12)$$

Here, q_{m_1,m_2} is a 2×2 matrix given by the double integral of the joint energy distribution $C(E_1, E_2)$ of two photons with energies E_1 and E_2 in the energy domain ϵ. The moment orders (m_1, m_2) control the weights that E_1 and E_2 apply to $C(E_1, E_2)$. The coefficients σ_0 to σ_3 originated from the work of Efthimiou et al., representing c_0 to c_3 along individual or small groups of LORs. a_m and b_m were the first raw moments of the non-scatter and scatter PDF, respectively, given by (19):

$$a_m = \int_{E_{min}}^{E_{max}} E^m \, U(E) \, dE \qquad b_m = \int_{E_{min}}^{E_{max}} E^m \, S(E) \, dE$$

When m equals 0, the energy term does not scale $U(E)$ or $S(E)$, and the normalised PDF has an area under the curve equal to 1, where:

$$a_0 = \int_{E_{min}}^{E_{max}} E^0 \, U(E) \, dE = 1 \qquad b_0 = \int_{E_{min}}^{E_{max}} E^0 \, S(E) \, dE = 1$$

$$a_1 = \int_{E_{min}}^{E_{max}} E^1 \, U(E) \, dE \qquad b_1 = \int_{E_{min}}^{E_{max}} E^1 \, S(E) \, dE$$

Therefore, in equation 12, the elements of the matrix q_{m_1,m_2} are given by:

$$\mathrm{q}_{0,0} = \sigma_0 \mathrm{a}_0 \mathrm{a}_0 + \sigma_0 a_0 b_0 + \sigma_0 b_0 a_0 + \sigma_3 b_0 b_0 = \sigma_0 + \sigma_1 + \sigma_2 + \sigma_3 \quad (13)$$

$$q_{0,1} = \sigma_0 a_0 a_1 + \sigma_0 a_0 b_1 + \sigma_0 b_0 a_1 + \sigma_3 b_0 b_1 = \sigma_0 a_1 + \sigma_1 b_1 + \sigma_2 a_1 + \sigma_3 b_1 \quad (14)$$

$$q_{1,0} = \sigma_0 a_1 a_0 + \sigma_0 a_1 b_0 + \sigma_0 b_1 a_0 + \sigma_3 b_1 b_0 = \sigma_0 a_1 + \sigma_1 a_1 + \sigma_2 b_1 + \sigma_3 b_1 \quad (15)$$

$$q_{1,1} = \sigma_0 a_1 a_1 + \sigma_0 a_1 b_1 + \sigma_0 b_1 a_1 + \sigma_3 b_1 b_1 \quad (16)$$

In practice, the estimated $U(E)$ and $S(E)$ spanned across discrete energy windows within $\hat{q}_{m_1,m_2}$ instead of continuous integrals, so equating $\hat{q}_{m_1,m_2}$ to q_{m_1,m_2} substituted the integral with finite sums over the recorded energies, resulting in a matrix equation:

$$\begin{bmatrix} \hat{q}_{0,0} \\ \hat{q}_{0,1} \\ \hat{q}_{1,0} \\ \hat{q}_{1,1} \end{bmatrix} = M \times \begin{bmatrix} \sigma_0 \\ \sigma_1 \\ \sigma_2 \\ \sigma_3 \end{bmatrix}$$

where:

$$\mathrm{M} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ a_1 & b_1 & a_1 & b_1 \\ a_1 & a_1 & b_1 & b_1 \\ a_1 \times a_1 & a_1 \times b_1 & b_1 \times a_1 & b_1 \times b_1 \end{bmatrix}$$

With discrete energy windows, the values of a_1 and b_1 were approximated as:

$$a_1 \approx \sum_i E_i \, U(E_i) \Delta E \quad (17)$$

$$b_1 \approx \sum_i E_i \, S(E_i) \Delta E \quad (18)$$

The non-scatter PDF in each energy window was multiplied by the window width $\Delta \mathrm{E}$ and the midpoint of that energy bin E_i. The scaled non-scatter PDF values were summed to produce a_1 present in equations 14–16. Based on equation 17, a_1 was calculated with a loop that iterated through all elements of $U(E)$ and multiplied each by the respective E_i and the window width. Similarly, b_1 in equation 18 was obtained by summing the scaled scatter PDF across all energy bins.

In each sinogram bin, solving equations 13–16 involved finding the inverse of matrix M:

$$\begin{bmatrix} \sigma_0 \\ \sigma_1 \\ \sigma_2 \\ \sigma_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ a_1 & b_1 & a_1 & b_1 \\ a_1 & a_1 & b_1 & b_1 \\ a_1 \times a_1 & a_1 \times b_1 & b_1 \times a_1 & b_1 \times b_1 \end{bmatrix}^{-1} \times \begin{bmatrix} \hat{q}_{0,0} \\ \hat{q}_{0,1} \\ \hat{q}_{1,0} \\ \hat{q}_{1,1} \end{bmatrix} \quad (19)$$

The inverse of the matrix M was calculated using `Numpy.linalg.inv`.

$\hat{q}_{m_1,m_2}$ was initiated as four empty 4D arrays with the same dimensions as the sinograms stored in `.hs` files. With a double `for` loop, the ground truth data were read into a sinogram with `AcquisitionData` to create $C(E_1, E_2)$ in each sinogram bin, where $\hat{q}_{m_1,m_2}$ was calculated as described above. The four arrays of $\hat{q}_{m_1,m_2}$ were stacked into a 5D tensor q.

Equation 19 was implemented with `np.tensordot(M_inv, q, axes=1)`, which conceptually multiplies the inverse of M by every slice of q along its first dimension—in this case, the four arrays of $\hat{q}_{m_1,m_2}$. A five-dimensional matrix was produced as the solution of this matrix equation, where the four values of σ were stacked. σ_1 to σ_3 were combined into the total coefficients of the scatter component. The mean values of the scatter component and the non-scatter component σ_0 across all 12 sinogram bins were displayed using `show_2D_array`. The sinograms from the ground truth were also visualised: the 12 non-scatter sinograms over all energy bins were read from `blue0_pink0` and displayed as the mean across all 12 sinogram bins. Similarly, the sinograms over all energy bins in `blue0_pink1`, `blue1_pink0`, and `blue1_pink1` were aggregated and displayed as the mean value across all sinogram bins.

Finally, the average values across all sinogram bins at view 16 were selected as a representative slice at the midpoint of the vertical axis to visualise the difference between our MoM estimation and the SimSET simulation. The curves were plotted by `plt.plot`.

4. Results and Discussion

4.1 Single Scatter Simulation

The activity and attenuation images in Fig. 3.2 were input into the SSS algorithm. The estimated SSS scatter sinogram was visualised as an array, as shown in Fig. 4.1. This scatter sinogram would be subtracted from the non-scatter-corrected data prior to the reconstruction process.

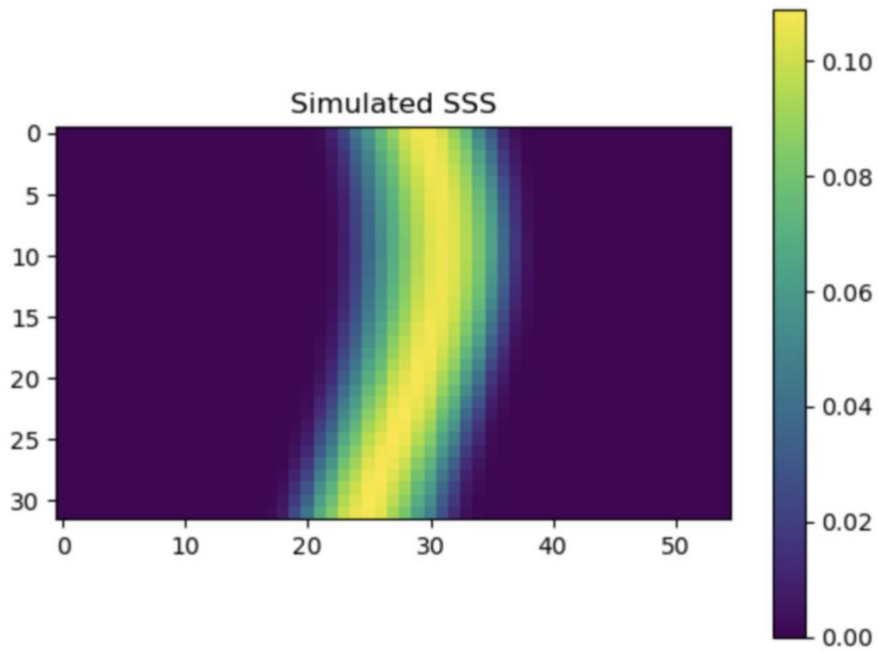

Fig. 4.1 The STIR-generated SSS scatter sinogram viewed at segment 0, axial bin 6. The sinogram of all LORs in axial bin 6 was displayed with 55 tangential bins on the horizontal-axis and 32 views on the vertical axis. The centre of the image shows a region of relatively higher scatter, with values greater than 0.1. This bright central band gradually decreases towards zero on both sides of the horizontal axis, spanning approximately eight tangential bins.

4.2 2D Histograms of Detected Events

Fig. 4.2 shows the distribution of weights for an attenuation cylinder with a radius of 150 mm and an activity cylinder with a radius of 24.22 mm. The weights of detected coincidence events from SimSET were binned into ten energy windows at two detectors (vertical and horizontal axes). The scale of the images was set to 0–3,100,000. Each pixel contains the summed weight of events (39) where one photon's energy was deposited in row x of detector 1 and the other photon's energy in column y of detector 2. The coordinates of a pixel are represented by (x, y).

The non-scatter histogram displays a rough Gaussian distribution centred around pixel $(5,5)$ that corresponds to 500–520 keV, which contains the expected energy of unscattered photons (511 keV). The peak value of weights of the histogram at $(5,5)$ is around 2.44×10^6. In contrast, the peak value in the scatter histogram Fig. 4.2B decreases to 7.00×10^5, but lower energies (400–480 keV) contain more weights. The weights recorded in windows 1–4 of both E_1 and E_2 are higher compared with Fig. 4.2A. Fig. 4.2C is the sum of Fig. 4.2A and 4.2B, displaying a very bright pixel at $(5,5)$ with a value of 3.07×10^6 and relatively higher weights in windows 1–4.

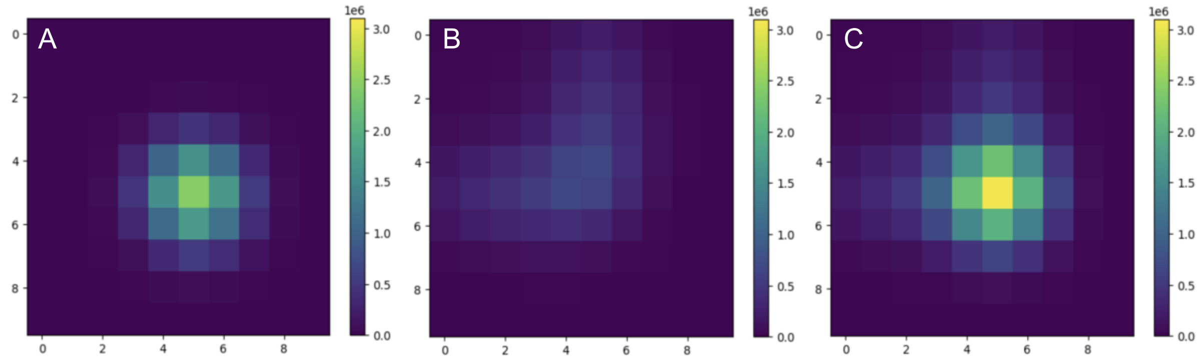

Fig. 4.2 The 2D histograms of the weights for detected coincidence events

Fig. 4.2A. The histogram of weights of unscattered photons. Starting from window 0, pixel $(5,5)$ has the highest summed weight. The weights in pixels around the peak are roughly symmetric. Windows 0–2 and 8–9 have lower weights.

Fig. 4.2B. The pixels with the highest summed weight are $(5,4)$ and $(4,5)$. Windows 1–4, however, are lighter in colour compared with Fig. 4.2A. Using the same intensity scale, the peak weight value of Fig. 4.2B is significantly lower than the peak value of Fig. 4.2A.

Fig. 4.2C. The total weight histogram was generated by summing Fig. 4.2A and Fig. 4.2B. The peak value of this histogram is located at $(5,5)$.

Converting these 2D histograms into 1D histograms, as described in Section 3.5, generated the spectra of weights across ten energy windows of E_1. As shown in Fig. 4.3, the three curves demonstrate the characteristic shapes introduced in Fig. 2.4. The non-scatter spectrum in light blue exhibits a peak at window 5 where 511 keV is located, forming a rough Gaussian distribution. The weight values in both high-energy windows (8–9) and low-energy windows (0–2) are small. The spectrum of the scatter photons in green, in contrast, remains greater than the non-scatter spectrum in windows 0–3, where scattered, low-energy photons are present. In particular, in windows 0–2, the scatter spectrum is close to the total spectrum.

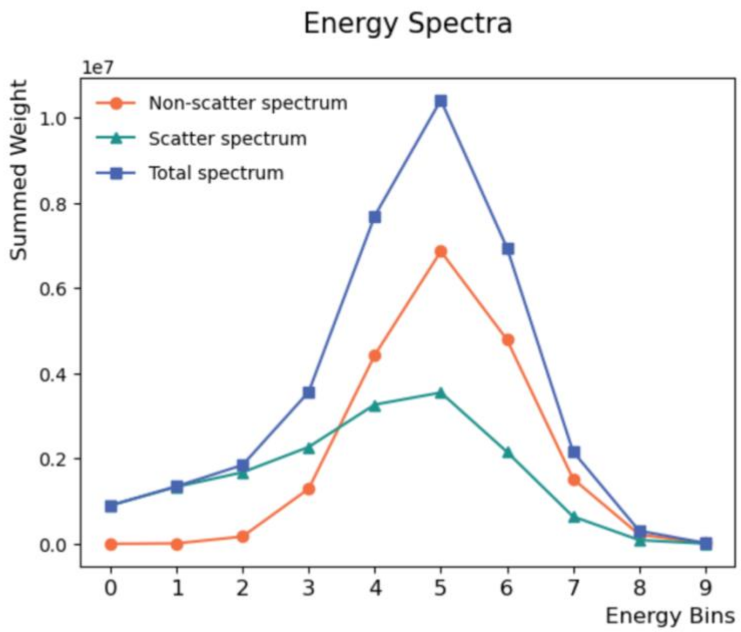

Fig. 4.3 The 1D spectra of simulated photons.

The spectrum of unscattered photons (green) shows a distinctive peak at window 5, whereas the scatter spectrum (orange) has a lower peak value but remains greater than the non-scatter spectrum in windows 0–3. Together, these two spectra form the total spectrum (blue).

4.3 Non-scatter and Scatter PDF

$U(E)$ was calculated with equation 5:

Table 4.1 The non-scatter PDF

Energy window	$U(E)$ ($\times 10^{-7}$)
0	3.63
1	173.36
2	3536.42
3	30851.71
4	115102.94
5	183647.81
6	125307.54
7	36564.59
8	4562.85
9	243.50

The 1D arrays of `blue0_pink0` and `blue0_pink1` were combined to form the curve of the non-scatter component as described in Section 3.5. The normalised non-scatter component and $U(E)$ were plotted in Fig. 4.4. $U(E)$ in orange aligns very closely with the normalised non-scatter component in green, where both curves display a peak at window 5 (500–520 keV) as expected. Three small gaps between the curves are present in windows 3, 5, and 7. In window 5 where the peaks are located, $U(E)$ is slightly greater than the non-scatter component. This is explained by the fact that our estimation slightly overestimates the non-scatter component. In contrast, the non-scatter component is slightly greater than $U(E)$ in windows 3 and 7. In all other windows, the two curves are closely matched with no clearly distinguishable differences.

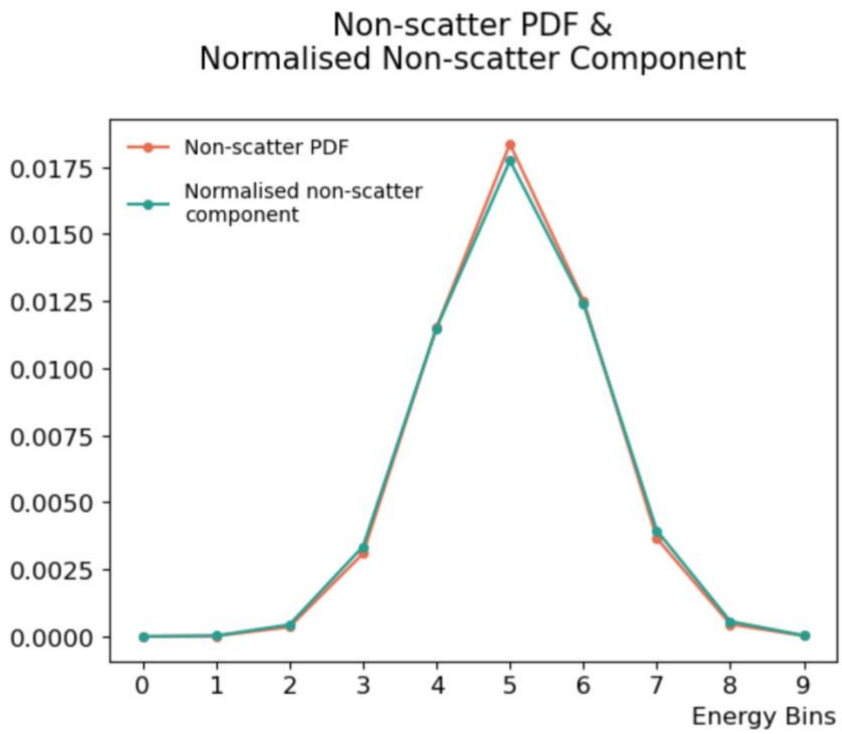

Fig. 4.4 Validation the estimated non-scatter PDF against the ground truth. $U(E)$ is in orange, whereas the ground truth of unscattered photons in `blue0_pink0_1D + blue0_pink1_1D` is shown by the green curve. Both curves follow a rough Gaussian distribution, with the orange curve slightly higher than the green curve at the peak.

The summed weight in windows 6–9 (520–600 keV) of the total energy spectrum was obtained, where few scattered photons were present. Using equations 7 and 8 in Section 3.5, directly scaling the non-scatter PDF to the order of magnitude of the total energy spectrum within this range resulted in a scaling factor of 5.66×10^8, which represents the total contribution of unscattered photons c_0'. This value is about 6% greater than the value of $(c_0 + c_1)$, which is equal to 5.34×10^8. From equation 10, subtracting the scaled non-scatter component from the total energy spectrum gives $(c_2 + c_3) \times S(E)$. However, this subtraction produced a negative value of the scaled scatter PDF in window 6.

Table 4.2 The scaled scatter PDF

Energy window	Scaled $S(E)$
0	899850.20
1	1338924.30
2	1647565.94
3	1815072.55
4	1176722.58
5	13673.05
6	-146357.02
7	93749.32
8	46903.61
9	5704.09

The negative value was capped at zero with `PDF_scatter_scaled = np.maximum(PDF_scatter_scaled, 0)`, and the resulting scaled $S(E)$ was normalised to obtain $S(E)$. The presence of the negative value after subtraction indicates some imperfections in the Gaussian model and the determination of the non-scatter component c_0', which are discussed in more detail on the next page. The 1D arrays of `blue1_pink0` and `blue1_pink1` were combined to form the scatter component of the ground truth, which was then normalised and visualised in order to validate our estimated $S(E)$, as shown in Fig. 4.5.

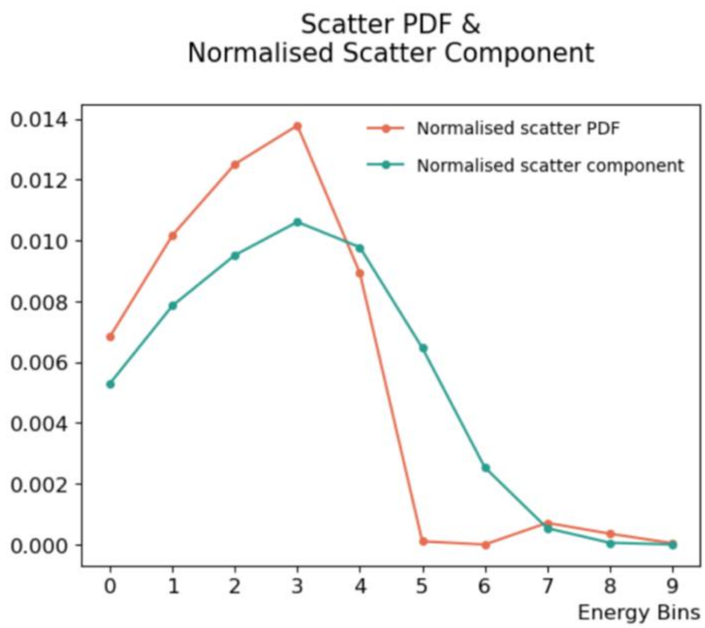

Fig. 4.5. Validation of the estimated scatter PDF with the normalised scatter component.

The normalised $S(E)$ is plotted in orange, while the normalised 1D array of the scatter component of the ground truth is shown in light green. Both curves display a peak at energy window 3, where $S(E)$ is greater than the peak value of the ground truth. Large gaps are present in windows 0–3 and 5–6.

Unlike $U(E)$, $S(E)$ does not closely align with the scatter component. In windows 0–3, the scatter PDF remains higher than the scatter component, while the values dramatically drop from window 4 until a negative value in window 6 (capped at zero). At high energy levels (windows 7–9) where scattered photons are few, the two curves are well aligned.

Our finding corroborates that one of the key limitations of this EBSE approach is its inability accurately quantify the contribution of the non-scatter component as a scaling factor of $U(E)$. Specifically, the value of c_0' was not close enough to the ideal value $(c_0 + c_1)$ as described above. While the Gaussian function captures the photopeak, it omits physical effects such as partial energy deposition and the influence of energy binning (19). Consequently, after subtraction, a significant discrepancy between the shapes of the estimated $S(E)$ and the simulated ground truth was observed. More importantly, a negative value of $S(E)$ was found in window 6 after subtraction, indicating a negative probability, which is mathematically invalid. The negative value of $S(E)$ was capped at zero before proceeding to MoM, leading to biased MoM-generated scatter sinograms.

The estimated $U(E)$ and $S(E)$ were input into the calculation of a_1 and b_1 in MoM, from which the scatter sinogram in each sinogram bin was generated.

4.4 Estimation of Scatter Sinograms Using MoM

The value of a_1 was computed by equation 17 at 510.99 keV, which is very close to the mean value at which $U(E)$ is centred; b_1 is approximately 485.60 keV, indicating that the energy of scattered photons is lower. Since the value of a_1 is extremely close to the initial emission energy of γ-photons, the EBSE-generated sinogram largely depends on the value of b_1.

The scatter sinograms generated by MoM are shown in Fig. 4.6. The midpoint of the vertical axis (view 16) was extracted from each sinogram as a 1D visualisation, shown in Fig. 4.7.

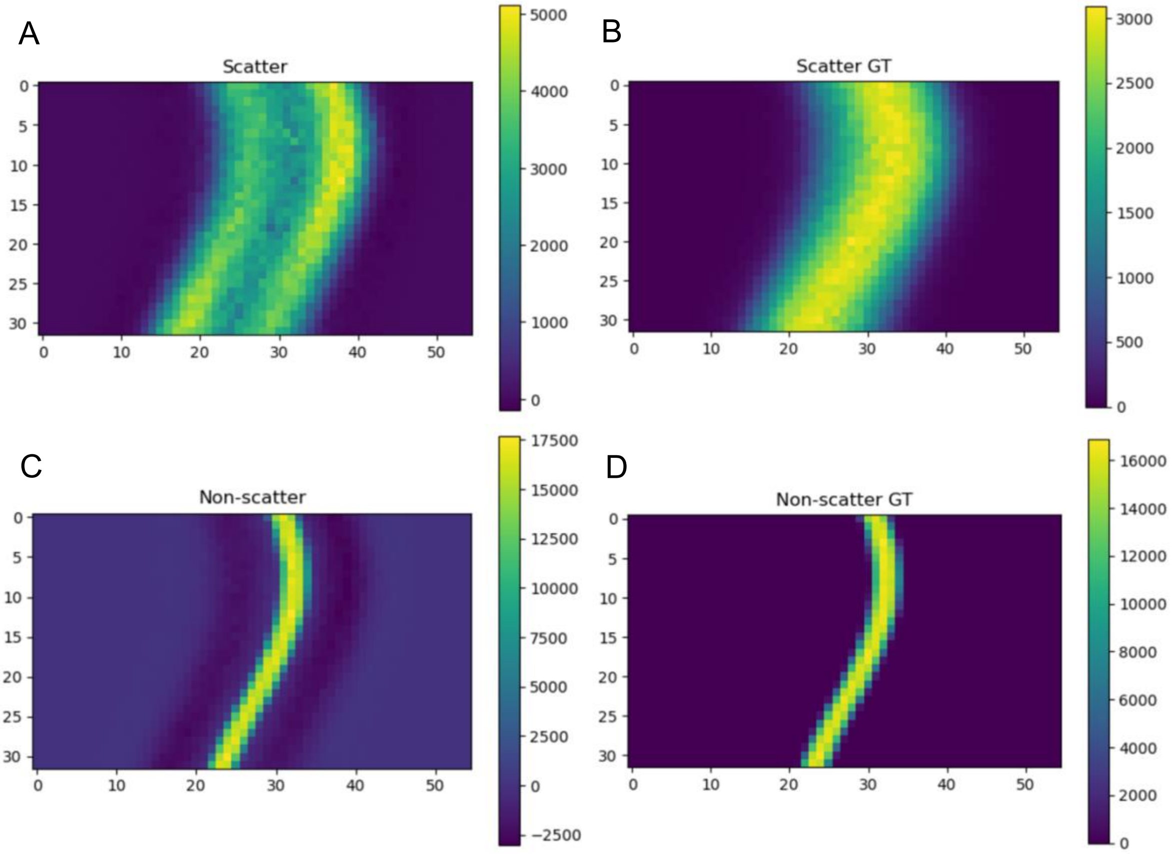

Fig. 4.6 The MoM-generated sinograms and the ground truth.

Fig. 4.6A and 4.6B show the scatter sinograms generated by MoM and read from the ground truth, respectively.

Fig. 4.6C and 4.6D show the sinograms for the unscattered photons.

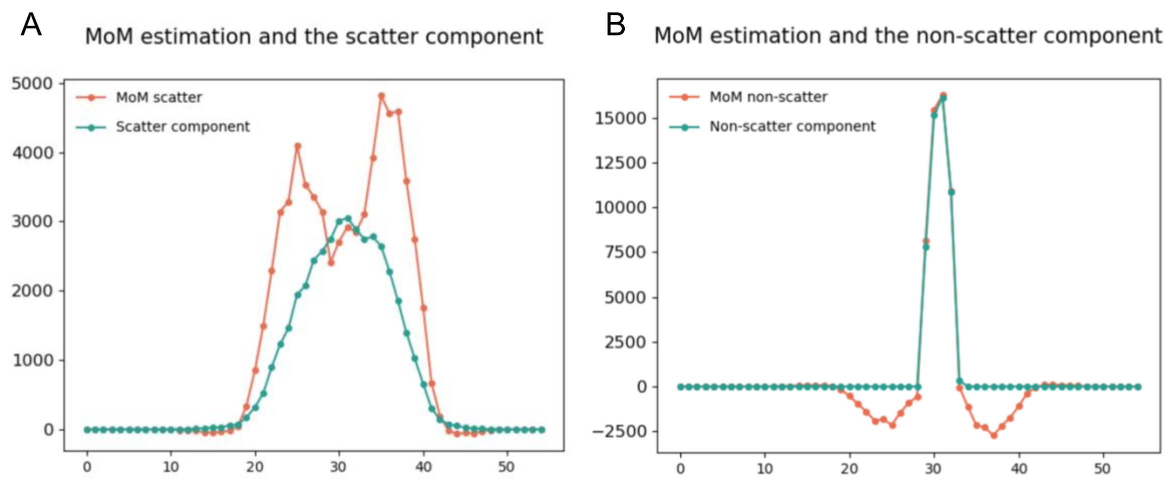

Fig. 4.7 View 16 of sinograms across 55 transaxial bins.

A horizontal line at view 16 of each sinogram was extracted and visualised.

Fig. 4.7A displays the MoM estimation of scatter events (orange) and the scatter component from the ground truth (green).

Fig. 4.7B illustrates the MoM estimation of non-scatter events (orange) and the non-scatter component from the ground truth (green).

The general shape of the MoM-generated sinogram (Fig. 4.6A) is similar to the ground truth in Fig. 4.6B and the SSS sinogram in Fig. 4.1, but the values and patterns differ: the MoM estimation exhibits a band of lower values in the middle, coupled with bands of higher values on either side; in contrast, the ground truth sinogram displays the reversed pattern, with the central region containing higher signal. The central regions of the respective non-scatter sinograms in Fig. 4.6C and 4.6D are very closely aligned, where both display an identical central band with high values. However, Fig. 4.6C presents two bands of negative values on either side of the central band, which are absent in Fig. 4.6D.

In Fig. 4.7A, around the peak region of the scatter component, the MoM estimation slightly concaves and does not deviate greatly from the ground truth, but on both sides of the peak region, the estimation models two higher peaks, leaving two large gaps. In contrast, the two curves in Fig. 4.7B are closely aligned at the peak region around bin 30, but the MoM estimation has two distinctive grooves on either side of the peak, where the values are negative, whereas the ground truth remains close to zero apart from the peak region.

Fig. 4.6 and Fig. 4.7 collectively show that our estimation provides accurate modelling of the unscattered photons at the signal peak, but the modelling of scattered photons is only roughly aligned with the simulated ground truth. More importantly, the two bands with negative values near the non-scatter events on Fig. 4.6C were not correctly modelled.

The LORs located at some distance from the emission source do not contain non-scatter events, so solving the 4×4 system of equations can lead to instability. The solution inflates the scatter component and assigns negative values to the absent non-scatter component (19), resulting in two peripheral bands with negative signal values in Fig. 4.6C. This mathematical artefact also explains the presence of two peripheral bands with higher signal than the central region of the sinogram in Fig. 4.6A. To overcome this limitation, Efthimiou et al. proposed a minimisation algorithm that removes negative values from the non-scatter component and re-solves the equation system, obtaining more stable and accurate estimates (19).

5. Conclusion and Future Directions

This work evaluates the method of Efthimiou et al. for an energy-based scatter estimation approach in positron emission tomography as an alternative to single scatter simulation (19). The contributions of unscattered and scattered photons were estimated based on the energy information of simulated photons. The energy distribution of the non-scatter component was fitted with a Gaussian function, and it aligned reasonably well with the Monte Carlo-simulated ground truth in the central region, although two bands of negative values were present on either side. However, the scatter component was not modelled with sufficient accuracy, and the estimated sinograms deviated from the ground truth, especially in regions adjacent to the photopeak. Overall, this study tested the methodology and workflow one approach of EBSE and demonstrated that it is a viable, realistic, and data-driven scatter correction approach that avoids the reliance of a transmission scan and the tail-fitting process required in SSS.

Future work should focus on improving the accuracy of estimating c_0', therefore refining the shape of $S(E)$. Since a Gaussian function cannot approximate c_0' with sufficient accuracy, several non-Gaussian approaches have been proposed. An empirically measured photopeak from a point source with minimised scattering can provide a better fit of the non-scatter PDF than a Gaussian function (19). A Monte Carlo simulation that separates scattered and unscattered photons enables purely simulation-driven characterisation (30), while an iterative expectation–maximisation approach can be implemented to determine the coefficients in equation 2 (20). Other simulation-based alternatives include directly separating non-scatter and scatter events during reconstruction based on pre-computed reference spectra (17). More recently, this concept has been extended to account for the difference in energy distribution between LORs at different spatial coordinates, tailoring non-scatter PDF of each LOR within the FOV based on its location (5).

Additionally, a minimisation algorithm that removes the negative bands in the non-scatter histogram was proposed by Efthimiou et al. (19). The implementation of this algorithm would yield a non-scatter sinogram that is closer to the ground truth. Moreover, performing a least-squares fit is an alternative approach to MoM (47). By minimising the sum of squared residuals between the measured full energy histogram and the linear combination of the four scenarios across all energy bins, the combination that best explains the data is produced. By aggregating the whole energy spectrum instead of using only four moments, a least-squares fit provides greater robustness and reduces random noise (48).

Image reconstruction with EBSE-corrected sinograms would be the next step of validation. Reconstruction can be performed using 3D list-mode ordered subsets expectation maximization (LM-OSEM), an iterative statistical algorithm. Since scatter events have a low frequency of spatial distribution, the sinograms in practice can be very noisy. Down-sampling the scanner geometry is a widely used denoising technique, producing sinograms with less noise that can then be up-sampled by full 3D interpolation (16, 19, 20). More recent advancements introduced deep learning-based noise reduction techniques that effectively remove noise in sinograms prior to reconstruction (49).

Although EBSE can be performed within a fairly short period for timely scatter correction, further improvements in computational efficiency remains in development. A recent study conducted by Pour et al. proposed a negative maximum likelihood algorithm that dramatically accelerates the process, where the estimate converges more quickly to a stable output as the number of simulated photons increases (5). This algorithm is likely to be extremely valuable in the foreseeable future, where EBSE needs to be evaluated with more complex phantoms and applied to real-world patient data (19, 20). The increased complexity of image volumes and the introduction of random coincidences can significantly raise the computational load. Eventually, the application of EBSE must undergo clinical validation to become a robust and timely scatter correction technique for clinical environments.

6. References

1. Zaidi H, Karakatsanis N. Towards enhanced PET quantification in clinical oncology. Br J Radiol. 2018;91(1081):20170508.
2. Ovtchinnikov E, Brown R, Kolbitsch C, Pasca E, da Costa-Luis C, Gillman AG, et al. SIRF: Synergistic Image Reconstruction Framework. Computer Physics Communications. 2020;249:107087.
3. Paredes-Pacheco J, López-González FJ, Silva-Rodríguez J, Efthimiou N, Niñerola-Baizán A, Ruibal Á, et al. SimPET-An open online platform for the Monte Carlo simulation of realistic brain PET data. Validation for (18) F-FDG scans. Med Phys. 2021;48(5):2482-93.
4. Hussain D, Abbas N, Khan J. Recent Breakthroughs in PET-CT Multimodality Imaging: Innovations and Clinical Impact. Bioengineering (Basel). 2024;11(12).
5. Pour SAZ, Rezaei A, Jansen F, Baete K, Schramm G, Nuyts J. Improved energy-based scatter estimation by incorporating local energy spectra and accelerating the parametric fitting. arXiv preprint arXiv:250415969. 2025.
6. Rong J, Haider A, Jeppesen TE, Josephson L, Liang SH. Radiochemistry for positron emission tomography. Nature Communications. 2023;14(1):3257.
7. Lamarca A, Barriuso J, Chander A, McNamara MG, Hubner RA, D OR, et al. (18)F-fluorodeoxyglucose positron emission tomography ((18)FDG-PET) for patients with biliary tract cancer: Systematic review and meta-analysis. J Hepatol. 2019;71(1):115-29.
8. Bousselham A. FPGA based data acquistion system and digital pulse processing for PET and SPECT2006.
9. Surti S. Update on Time-of-Flight PET Imaging. Journal of Nuclear Medicine. 2015;56(1):98.
10. Fahey F. Data acquisition in PET imaging. Journal of nuclear medicine technology. 2002;30:39-49.
11. Tong S, Alessio AM, Kinahan PE. Image reconstruction for PET/CT scanners: past achievements and future challenges. Imaging Med. 2010;2(5):529-45.
12. Lindén J, Teuho J, Teräs M, Klén R. Evaluation of three methods for delineation and attenuation estimation of the sinus region in MR-based attenuation correction for brain PET-MR imaging. BMC Med Imaging. 2022;22(1):48.
13. Salvadori J, Labour J, Odille F, Marie PY, Badel JN, Imbert L, et al. Monte Carlo simulation of digital photon counting PET. EJNMMI Phys. 2020;7(1):23.
14. Efthimiou N, Emond E, Wadhwa P, Cawthorne C, Tsoumpas C, Thielemans K. Implementation and validation of time-of-flight PET image reconstruction module for listmode and sinogram projection data in the STIR library. Physics in Medicine & Biology. 2019;64(3):035004.
15. Wells RG, deKemp RA. Does time-of-flight improve image quality in the heart? Journal of Nuclear Cardiology. 2019;26(2):413-6.
16. Nassiri M, Hissoiny S, Carrier J-F, Després P. Fast GPU-based computation of the sensitivity matrix for a PET list-mode OSEM algorithm. Physics in medicine and biology. 2012;57:6279-93.
17. Álvarez-Gómez JM, Santos-Blasco J, Moliner Martínez L, Rodríguez-Álvarez MJ. Fast Energy Dependent Scatter Correction for List-Mode PET Data. J Imaging. 2021;7(10).
18. Ollinger JM. Model-based scatter correction for fully 3D PET. Phys Med Biol. 1996;41(1):153-76.
19. Efthimiou N, Karp JS, Surti S. Data-driven, energy-based method for estimation of scattered events in positron emission tomography. Phys Med Biol. 2022;67(9).

20. Hamill JJ, Cabello J, Surti S, Karp JS. Energy-based scatter estimation in clinical PET. Med Phys. 2024;51(1):54-69.
21. Azhari H, Kennedy JA, Weiss N, Volokh L. Positron Emission Tomography (PET). In: Azhari H, Kennedy JA, Weiss N, Volokh L, editors. From Signals to Image: A Basic Course on Medical Imaging for Engineers. Cham: Springer International Publishing; 2020. p. 217-51.
22. Visvikis D, Griffiths D, Costa D, Bomanji J, Ell P. Clinical evaluation of 2D versus 3D whole-body PET image quality using a dedicated BGO PET scanner. European journal of nuclear medicine and molecular imaging. 2005;32:1050-6.
23. Lodge MA, Badawi RD, Gilbert R, Dibos PE, Line BR. Comparison of 2-Dimensional and 3-Dimensional Acquisition for 18F-FDG PET Oncology Studies Performed on an LSO-Based Scanner. Journal of Nuclear Medicine. 2006;47(1):23.
24. Faasse T. Positron Emission Tomography-Computed Tomography Data Acquisition and Image Management. In: Misciagna S, editor. Positron Emission Tomography - Recent Developments in Instrumentation, Research and Clinical Oncological Practice. Rijeka: IntechOpen; 2013.
25. Laurent B, Bousse A, Merlin T, Nekolla S, Visvikis D. PET scatter estimation using deep learning U-Net architecture. Physics in Medicine & Biology. 2023;68(6):065004.
26. Watson CC, Newport D, Casey ME, deKemp RA, Beanlands RS, Schmand M. Evaluation of simulation-based scatter correction for 3-D PET cardiac imaging. IEEE Transactions on Nuclear Science. 1997;44(1):90-7.
27. Watson CC, editor Extension of single scatter simulation to scatter correction of time-of-flight PET. IEEE Nuclear Science Symposium Conference Record, 2005; 2005 23-29 Oct. 2005.
28. Magota K, Numata N, Shinyama D, Katahata J, Munakata Y, Maniawski PJ, et al. Halo artifacts of indwelling urinary catheter by inaccurate scatter correction in (18)F-FDG PET/CT imaging: incidence, mechanism, and solutions. EJNMMI Phys. 2020;7(1):66.
29. Lingxiong S, Freifelder R, Karp JS. Triple energy window scatter correction technique in PET. IEEE Transactions on Medical Imaging. 1994;13(4):641-8.
30. Popescu LM, Lewitt RM, Matej S, Karp JS. PET energy-based scatter estimation and image reconstruction with energy-dependent corrections. Phys Med Biol. 2006;51(11):2919-37.
31. Grootoonk S, Spinks TJ, Sashin D, Spyrou NM, Jones T. Correction for scatter in 3D brain PET using a dual energy window method. Phys Med Biol. 1996;41(12):2757-74.
32. Bendriem B, Trebossen R, Frouin V, Syrota A, editors. A PET scatter correction using simultaneous acquisitions with low and high lower energy thresholds. 1993 IEEE Conference Record Nuclear Science Symposium and Medical Imaging Conference; 1993 31 Oct.-6 Nov. 1993.
33. Rezaei A, Schramm G, Willekens SMA, Delso G, Van Laere K, Nuyts J. A Quantitative Evaluation of Joint Activity and Attenuation Reconstruction in TOF PET/MR Brain Imaging. J Nucl Med. 2019;60(11):1649-55.
34. Thielemans K, Tsoumpas C, Mustafovic S, Beisel T, Aguiar P, Dikaios N, et al. STIR: software for tomographic image reconstruction release 2. Phys Med Biol. 2012;57(4):867-83.
35. Thielemans K. STIR Documentation 2024 [cited 2025 August 11]. Available from: https://stir.sourceforge.net/documentation/.
36. Brix G, Zaers J, Adam LE, Bellemann ME, Ostertag H, Trojan H, et al. Performance evaluation of a whole-body PET scanner using the NEMA protocol. National Electrical Manufacturers Association. J Nucl Med. 1997;38(10):1614-23.
37. Wienhard K, Dahlbom M, Eriksson L, Michel C, Bruckbauer T, Pietrzyk U, et al. The ECAT

EXACT HR: performance of a new high resolution positron scanner. J Comput Assist Tomogr. 1994;18(1):110-8.
38. University of Washington DoNM. Simulation System for Emission Tomography (SimSET) 1993 [updated 2017 Feb 1. 2.9.2:[Available from: https://depts.washington.edu/simset/html/simset_home.html.
39. Haynor DR, Harrison RL, Lewellen TK. The use of importance sampling techniques to improve the efficiency of photon tracking in emission tomography simulations. Medical Physics. 1991;18(5):990-1001.
40. Baum KG, Helguera M. Execution of the SimSET Monte Carlo PET/SPECT simulator in the condor distributed computing environment. J Digit Imaging. 2007;20 Suppl 1(Suppl 1):72-82.
41. Harrison RL, Vannoy SD, Haynor DR, Gillispie SB, Kaplan MS, Lewellen TK, editors. Preliminary Experience With The Photon History Generator Module Of A Public-domain Simulation System For Emission Tomography. 1993 IEEE Conference Record Nuclear Science Symposium and Medical Imaging Conference; 1993 31 Oct.-6 Nov. 1993.
42. Wicklin R. SAS [Internet]2022. Available from: https://blogs.sas.com/content/iml/2022/09/28/definition-of-moments.html?utm_source=chatgpt.com.
43. Anaconda. Installing Miniconda. [cited 2025 August 12th]. Available from: https://www.anaconda.com/docs/getting-started/miniconda/install#macos-2.
44. Thielemans K. Installing STIR with conda [updated March 18, 2025; cited 2025 August 11]. Available from: https://github.com/UCL/STIR/wiki/Installing-STIR-with-conda.
45. SyneRBI. SIRF Docker Image [cited 2025 August 11]. Available from: https://github.com/SyneRBI/SIRF-SuperBuild/blob/master/docker/README.md.
46. Rainio O, Hällilä J, Teuho J, Klén R. Methods for estimating full width at half maximum. Signal, Image and Video Processing. 2025;19(4):289.
47. Teuho J, Saunavaara V, Tolvanen T, Tuokkola T, Karlsson A, Tuisku J, et al. Quantitative Evaluation of 2 Scatter-Correction Techniques for ¹⁸F-FDG Brain PET/MRI in Regard to MR-Based Attenuation Correction. Journal of Nuclear Medicine. 2017;58(10):1691.
48. Thielemans K, Manjeshwar RM, Tsoumpas C, Jansen FP, editors. A new algorithm for scaling of PET scatter estimates using all coincidence events. 2007 IEEE Nuclear Science Symposium Conference Record; 2007 26 Oct.-3 Nov. 2007.
49. Jiang C, Liu M, Sun K, Shen D. End-to-end Triple-domain PET Enhancement: A Hybrid Denoising-and-reconstruction Framework for Reconstructing Standard-dose PET Images from Low-dose PET Sinograms2024.

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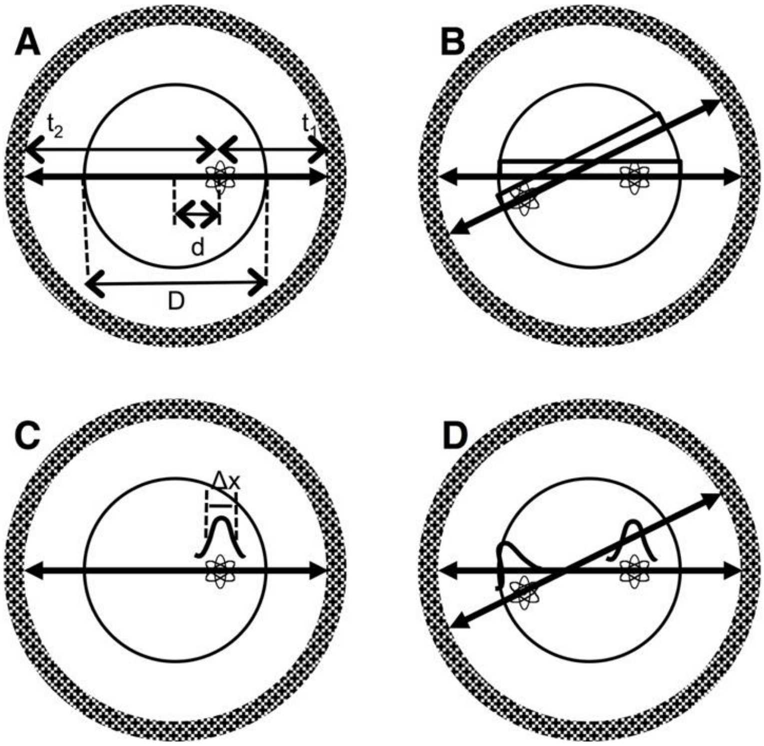
I also give heartfelt thanks to my friends and family that supported me throughout this rewarding journey. The encouragement from Mia, Erwin, James, and Abigail has been a constant source of motivation and relief during the challenging days.

8. Appendix

https://zenodo.org/records/17014917?token=eyJhbGciOiJIUzUxMiJ9.eyJpZCI6ImU2ODU4MTI5LWEzNzMtNDhhZC1iZWY0LTVhZTYxODBkMDQ1MCIsImRhdGEiOnt9LCJyYW5kb20iOiJiOTY2M2U3YmQ1YmM4ODAyMDNkYjZjN2JkODAxYjhlZSJ9.HEcXy4Kp7qR-WbZGoc7X0SKCAZW0o6D32LIdH79-IqJywsMFBXk9W56Q-c0YeNFJ-uTfvu0o4IhFKfFdj-5sCg

This Zenodo link contains the `.ipynb` file used in STIR and a folder `2.9.2`, which the scripts of SimSET and input files are located at. The output files of SimSET were contained in `8.10`, and the folder `STIR` contains the utilities needed in STIR and SimSET.

The supplementary figures begin on the next page.

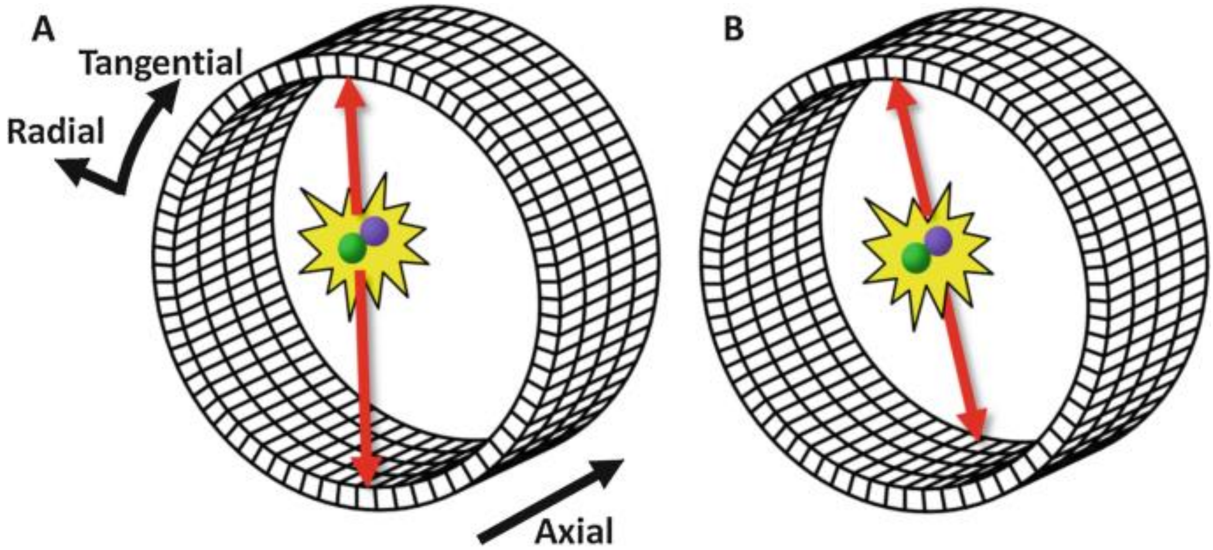


Supplementary Fig. 1. Noise correlation in a non-TOF PET scanner (9).

Sup. Fig. 1A. Within an object with a diameter of D, the emission point is located at distance d from the centre of the scanner. The two photons in coincidence reach the detector ring with respective arrival times t_1 and t_2.

Sup. Fig. 1B. Without precise TOF measurement, each of the two emission points is located with uniform probability along the respective LOR. Noise correlation is therefore present at the intersection of these two LORs.

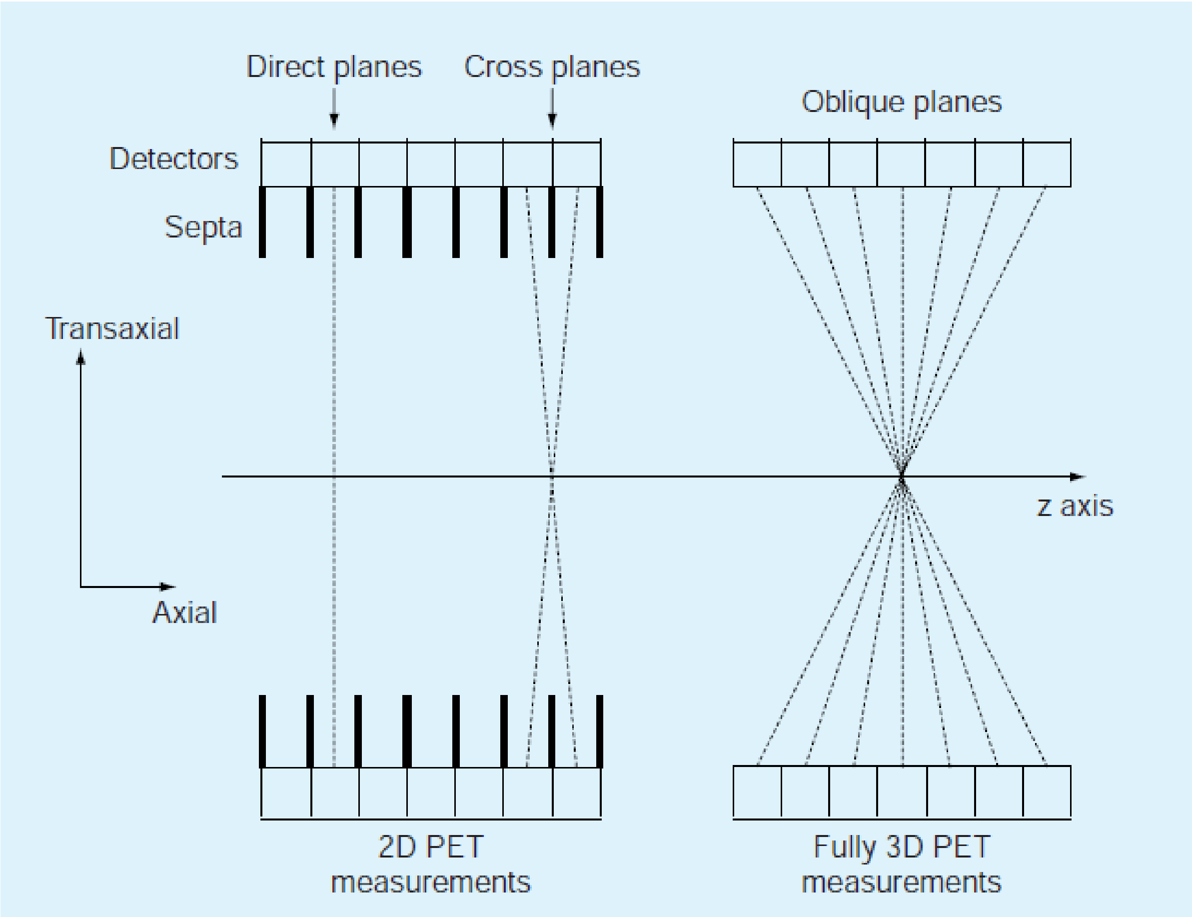
Sup. Fig. 1C and 1D. TOF measurement determines the locations of emission points and reduces unwanted noise correlation.



Supplementary Fig. 2. The directions in 3D PET (21).

Sup. Fig. 2A shows the directions in the ring configuration. The z-axis refers to the axial direction. The radial and tangential axes together form the transaxial plane, which is perpendicular to the axial direction.

Sup. Fig. 2B is an illustration of 3D PET detecting a coincidence event with two separate rings.



Supplementary Fig. 3. A figurative illustration of 2D and 3D PET in side view (11).

In 2D PET (left), septa of tungsten or lead are placed between detector rings. Only coincidences detected within a plane or adjacent planes can be recorded to reduce the dead time of scintillation crystals.

The improvement in crystal and detector technologies enabled the removal of septa. All coincidence events in oblique planes can be detected by 3D PET, resulting in significantly improved sensitivity.